

MULTIVARIABLE FORMAL CATEGORY THEORY

PATRICK SCHULTZ AND MICHAEL SHULMAN

CONTENTS

1. INTRODUCTION

We propose a categorical structure designed for doing “multivariable formal category theory”, including particularly contravariant functors and extraordinary natural transformations, and hence the abstract study of monoidal objects, enriched objects, multivariable adjunctions, closed objects, rigid objects, and so on. This is analogous to how proarrow equipments [?] and Yoneda structures [?] provide abstract contexts for doing one-variable formal category theory. By analogy with the word “equipment”, we call our structure a **kit**.

There does already exist one structure in which one can do multivariable formal category theory. As observed by Day and Street [?], the opposite of a category is its dual in the monoidal bicategory of profunctors, making the latter “compact closed” (also called “rigid” or “autonomous”); and inside a compact closed bicategory one can define structures of multivariable category theory. Of course, the bicategory of profunctors does not quite know which profunctors are representable by functors (the left adjoint profunctors are “representable up to Cauchy completion”), but this can be remedied by enhancing it to a “compact closed proarrow equipment”.

The differences between a kit and a compact closed equipment, and hence the reasons for defining the former, are:

- (i) Compact closed equipments have not actually been defined yet either, and it’s not necessarily *a priori* obvious how much coherence should be required between the “functor arrows” and the compact closed structure. (As observed in [?], even for just the one-variable “opposite category” operation there are nontrivial coherence questions.) The definition of a kit is “more obviously complete”. Below, we will actually also define compact closed equipments and prove that any such gives rise to a kit, justifying that definition.
- (ii) A kit is “fully virtual” in the sense of [?]; that is, all of its structure is multicategorical. This means that rather than including *operations* such as profunctor composition, monoidal product, and dualization, a kit includes a basic notion of “morphism out of such things”, in the same way that a multicategory has no actual monoidal product but includes a basic notion of “multivariable morphism” that “would be a morphism out of a monoidal product if it existed”. This allows these operations, when they exist, to be characterized by universal properties, eliminating the question of what coherence axioms have to be imposed upon them as structure. It also allows us to describe examples where such operations do *not* exist. For instance,

- categories enriched over any monoidal category $\mathbf{V}$ form a kit, whether or not $\mathbf{V}$ has any limits or colimits; indeed $\mathbf{V}$ could be only a multicategory itself.
- (iii) Because of virtuality, it is much simpler to define sub-kits and quotient kits. In particular, this allows us to construct “homotopy kits” by first restricting to a subclass of well-behaved objects, then quotienting by a homotopy relation. This gives one approach to formal higher category theory using only 2-categorical machinery.
 - (iv) A kit eliminates the “two-sided bias” of a bicategory or an equipment. Two-sidedness is appropriate and necessary for one-variable category theory, since in examples such as categories enriched in a non-symmetric monoidal category (or in a bicategory) the only “modules” that can be defined are covariant in exactly one variable and contravariant in exactly one other variable. But for multivariable category theory, where a module can naturally be covariant in many variables and contravariant in many other variables, it feels artificial to divide these variables into a “domain” and a “codomain”, forcing us to pass back and forth across dualization equivalences when we want to compare a profunctor $A \rightrightarrows B \times C$ with a profunctor $C^{\text{op}} \times A \rightrightarrows B$, while intuitively (and in most examples) there is really only one notion involved, namely a functor $B^{\text{op}} \times C^{\text{op}} \times A \rightarrow \mathbf{Set}$.
 - (v) Finally, a kit furnishes a more natural semantics for the “directed type theory” of [?], because its classes of objects and morphisms correspond directly to the judgments of the latter. In particular, a kit admits a very familiar set-like “internal-language” in which to reason about its objects, analogous to the “abstract index notation” of tensor calculus. We will introduce this language informally here; for details see [?].

Part 1. Multivariable formal category theory

In this part we will describe the structure of a kit informally, and consider a number of examples of how to use that structure to formulate category-theoretic arguments abstractly. The formal definition of a kit is rather technically involved and occupies ??; but this part is intended to demonstrate that for nearly all practical applications it is not necessary to know the details of the formal definition, only to have faith that it exists and behaves as one would expect.

2. THE STRUCTURE OF A KIT

A kit is composed of four sorts of things, which we discuss in turn.

2.1. Cats. There is a class of objects, which we call **cats**, since they play the role of categories.

2.2. Functors. We generally call the morphisms between cats **functors**. This is in line with the fact that the morphisms between many different kinds of generalized categories — enriched categories, indexed categories, bicategories, ∞ -categories, etc. — are frequently called just “functors” without qualification.

The functors in a kit are functors of many variables, some covariant and some contravariant. More precisely, the domain of a functor is a finite list of cats, some of which are annotated with the formal symbol $(-)^{\circ}$ to indicate contravariance. (This does not yet represent an operation of “oppositization” on cats; it is just a formal marker for contravariance. Similarly, we do not in general assume a “tensor product”

of cats; as we will see later, these operations can be characterized by universal properties in terms of the multivariable functors.) The codomain of a functor, by contrast, is a single cat; thus we might have for instance $f : (A, B^\circ, C) \rightarrow D$. If the list contains only one cat, we generally omit the parentheses, writing $f : A \rightarrow B$.

Functors can be composed in a “multicategorical” way that keeps track of variance. For instance, if in addition to $f : (A, B^\circ, C) \rightarrow D$ we have $g : X \rightarrow A$ and $h : (Y^\circ, Z) \rightarrow B$ and $k : () \rightarrow C$ (yes, lists can have length zero), then there is a composite $f \circ (g, h, k) : (X, Y, Z^\circ) \rightarrow D$. Finally, like in a symmetric multicategory, we have symmetric group actions on the domains of functors, so that from $f : (A, B^\circ, C) \rightarrow D$ we can obtain a functor $(B^\circ, C, A) \rightarrow D$ and so on.

In particular, we have a category of cats and unary functors, which we denote $\mathbb{K}_1$. Later on we will enhance this to a 2-category.

2.3. Modules. Now we have a second class of objects, which we call **modules**; these represent presheaves, profunctors, and more general “functors into the base of enrichment”. Each module is indexed by a finite list of cats with variance (the same sort of list that appears as the domain of a functor). We write $\text{Mod}_{\mathbb{K}}(A, B^\circ, C)$ for the category (see below) of modules indexed by a list (A, B°, C) . Modules can be restricted along functors; thus if $M \in \text{Mod}_{\mathbb{K}}(A, B^\circ, C)$, and g, h, k are as above, we have $(g, h, k)^*M \in \text{Mod}_{\mathbb{K}}(X, Y, Z^\circ)$. We will also write $(g, h, k)^*M$ as $M(g, h, k)$. This restriction is strictly functorial with respect to functor composition (one could consider generalizations of a kit where it is only weakly functorial, but in most examples it is strict, and in other cases it can be modified to become so).

The symmetric groups also act on modules; thus if $M \in \text{Mod}_{\mathbb{K}}(A, B^\circ, C)$ we have a module $\sigma^*M \in \text{Mod}_{\mathbb{K}}(B^\circ, C, A)$ and so on. As with restriction, we assume for simplicity that this action is strict. Formally, the symmetric group action is combined with the restriction along functors, by representing the collection of modules as a functor defined on a category whose objects are lists of cats with variance and whose morphisms are lists of functors together with a permutation of their combined input.

2.4. Module morphisms. Finally we have module morphisms; this is where all the subtlety and difficulty lies. The simplest sort of morphism between modules is an “ordinary” one between modules indexed by the same list of cats. These form the morphisms of the categories such as $\text{Mod}_{\mathbb{K}}(A, B^\circ, C)$, and restriction along functors is of course functorial on these categories.

The next simplest sort of module morphism is a “multivariable” one that represents a potential “tensor product of modules” multicategorically, similarly to how the multivariable functors represent a potential tensor product of categories. Since our intended models include categories enriched over a non-cartesian base, the individual categories such as $\text{Mod}_{\mathbb{K}}(A, B^\circ, C)$ are not necessarily monoidal (that is, we can’t necessarily take a pointwise tensor product of presheaves), but we can expect “external” products such as $\text{Mod}_{\mathbb{K}}(A) \times \text{Mod}_{\mathbb{K}}(B) \rightarrow \text{Mod}_{\mathbb{K}}(A, B)$. Thus, representing these multicategorically we have multivariable module morphisms such as $(M_1, M_2, \dots, M_n) \rightarrow N$, where $M_i \in \text{Mod}_{\mathbb{K}}(\Psi_i)$ and $N \in \text{Mod}_{\mathbb{K}}(\Psi_1, \Psi_2, \dots, \Psi_n)$, where $(\Psi_1, \Psi_2, \dots, \Psi_n)$ indicates the concatenated list of cats with variance. As with functors, the symmetric groups act on the domains of such morphisms.

However, in order to really do multivariable category theory, we also need “extraordinary” natural transformations as in [?]. For instance, given $M \in \text{Mod}_{\mathbb{K}}(A, A^\circ, B)$

and $N \in \mathcal{Mod}_{\mathbb{K}}(C, C^\circ, B)$ we want to speak about a transformation $\alpha : M \rightarrow N$ that is ordinary natural in B and extraordinary natural in A and C on either side. In plain set-based category theory this means we have components $\alpha_{a,b,c} : M(a, a, b) \rightarrow N(c, c, b)$ such that the following naturality squares commute for all $f : a_1 \rightarrow a_2$, $g : b_1 \rightarrow b_2$, $h : c_1 \rightarrow c_2$, and a, b, c :

$$\begin{array}{ccc}
 M(a, a, b_1) & \xrightarrow{\alpha} & N(c, c, b_1) \\
 M(1,1,g) \downarrow & & \downarrow N(1,1,g) \\
 M(a, a, b_2) & \xrightarrow{\alpha} & N(c, c, b_2)
 \end{array}
 \qquad
 \begin{array}{ccc}
 M(a_1, a_2, b) & \xrightarrow{M(f,1,1)} & M(a_2, a_2, b) \\
 M(1,f,1) \downarrow & & \downarrow \alpha \\
 M(a_1, a_1, b) & \xrightarrow{\alpha} & N(c, c, b)
 \end{array}$$

$$\begin{array}{ccc}
 M(a, a, b) & \xrightarrow{\alpha} & N(c_2, c_2, b) \\
 \alpha \downarrow & & \downarrow N(1,h,1) \\
 N(c_1, c_1, b) & \xrightarrow{N(h,1,1)} & N(c_2, c_1, b)
 \end{array}$$

Thus, a kit includes a set of “extraordinary” module morphisms from M to N whenever $M \in \mathcal{Mod}_{\mathbb{K}}(\Psi_1, \Psi_1^\circ, \Psi_2)$ and $N \in \mathcal{Mod}_{\mathbb{K}}(\Psi_3, \Psi_3^\circ, \Psi_2)$. Here the Ψ_i are lists of cats with variance, and Ψ° denotes reversal of all the variances in Ψ . Of course, in general the cats we want to match up extraordinarily may not appear in exactly this order (e.g. we could have $M \in \mathcal{Mod}_{\mathbb{K}}(A, B, A^\circ)$ and $N \in \mathcal{Mod}_{\mathbb{K}}(B, C^\circ, C)$), but we can first act on the modules by a permutation.

The extraordinary morphisms, like the ordinary ones that form the arrows of the categories $\mathcal{Mod}_{\mathbb{K}}(\Psi)$, can be restricted along functors, in a functorial way. Specifically, given $M \in \mathcal{Mod}_{\mathbb{K}}(\Psi_1, \Psi_1^\circ, \Psi_2)$ and $N \in \mathcal{Mod}_{\mathbb{K}}(\Psi_3, \Psi_3^\circ, \Psi_2)$ as above, determining a type of extraordinary module morphism from M to N , then such module morphisms can be restricted along any lists of functors $f_i : \Psi'_i \rightarrow \Psi_i$ (for $i = 1, 2, 3$). Here f_i consists of a functor with codomain A for each cat A in Ψ_i , and Ψ'_i is the concatenated list of the domains of these functors, with the variance of each A in Ψ_i applied to the domain of the corresponding functor. Thus for instance if $g : (A, B^\circ) \rightarrow C$ and $h : D \rightarrow E$, we would have $(g, h) : (A, B^\circ, D^\circ) \rightarrow (C, E^\circ)$. Note that just as the domain and codomain of an ordinary module morphism have to be restricted along the same functor (since restriction defines functors $\mathcal{Mod}_{\mathbb{K}}(\Psi) \rightarrow \mathcal{Mod}_{\mathbb{K}}(\Psi')$), the two occurrences of a cat in the domain or codomain in an extraordinary morphism also have to be restricted along the same functor.

Finally, we can have multivariable extraordinary morphisms as well, where the domain is a list of modules rather than a single one. In this case we almost always want to permute the lists of cats appearing in the domain rather than simply concatenating them, so technically we have to incorporate a notion of permutation action on a list; in a moment we will introduce a more convenient notation for this.

The real complication comes when defining how extraordinary morphisms can be *composed*. The rules for how to compose extraordinary natural transformations were given in [?], but it seems to be quite involved to use these rules to define an abstract structure in a categorically respectable way. In ?? we will describe the rules informally; but first we digress to introduce a better notation for extraordinary morphisms.

3. ABSTRACT VARIABLE NOTATION

It turns out to be quite convenient to use a different notation for modules and their morphisms. Rather than writing

$$M \in \mathcal{Mod}_{\mathbb{K}}(A, B^\circ, C)$$

we allow ourselves to think of M as a functor that takes “values” on “variables” belonging to A, B, C , and write

$$(a : A), (b : B^\circ), (c : C) \vdash M(a, b, c) \in \mathcal{Mod}_{\mathbb{K}}.$$

Note that the cats in a kit are still just abstract objects; we are not asserting that they have “elements” in any sense. These “variables” are purely schematic. This is very much like the “abstract index notation” of tensor calculus that allows us to write T_c^{ab} for an element of a tensor space $V \otimes V \otimes V^*$ without needing to actually pick a basis of V to make a, b, c into actual numerical indices of components.

One reason this “abstract variable notation” is useful is when restricting modules along functors. In the previous example with M as above and

$$g : X \rightarrow A \quad h : (Y^\circ, Z) \rightarrow B \quad k : () \rightarrow C$$

instead of $(g, h, k)^*M$ or $M(g, h, k)$ we can write

$$(x : X), (y : Y), (z : Z^\circ) \vdash M(g(x), h(y, z), k) \in \mathcal{Mod}_{\mathbb{K}}$$

This indicates much more clearly how the functors g, h, k match up with the dependence of M on A, B°, C and how the resulting dependence on X, Y, Z° feeds into g, h, k . The strict functoriality of restriction with respect to functor composition means that this notation behaves as we expect on such composites; for instance, if we have

$$(c : C) \vdash M(c) \in \mathcal{Mod}_{\mathbb{K}} \quad g : B \rightarrow C \quad f : A \rightarrow B$$

then the three restrictions g^*M , f^*g^*M , and $(g \circ f)^*M$ are represented in this notation as

$$(b : B) \vdash M(g(b)) \in \mathcal{Mod}_{\mathbb{K}}$$

$$(a : A) \vdash M(g(f(a))) \in \mathcal{Mod}_{\mathbb{K}} \quad (a : A) \vdash M((g \circ f)(a)) \in \mathcal{Mod}_{\mathbb{K}}$$

where the second and third are, as we expect, equal.

Remark 3.1. To avoid confusion between functors and abstract variables, we emphasize that the symbol $\vdash$ always indicates the use of abstract variables, and in its absence we always revert back to the notation $M(g, h, k)$.

The second place that abstract variable notation is useful is when describing module morphisms, and especially extraordinary ones. We said above that extraordinary module morphisms go from M to N whenever $M \in \mathcal{Mod}_{\mathbb{K}}(\Psi_1, \Psi_1^\circ, \Psi_2)$ and $N \in \mathcal{Mod}_{\mathbb{K}}(\Psi_3, \Psi_3^\circ, \Psi_2)$. We also mentioned that we generally have to apply a permutation in order to put the lists of cats in exactly this form, such as if $M \in \mathcal{Mod}_{\mathbb{K}}(A, B, A^\circ)$ and $N \in \mathcal{Mod}_{\mathbb{K}}(B, C, C^\circ)$. In practice, instead of actually acting on M and N by symmetric groups, it is easier to just indicate how the cats should be matched up. We can do this with abstract variables as follows:

$$(a : A), (b : B), (c : C) \mid M(a, b, a) \vdash N(b, c, c)$$

Note that every variable appears exactly twice to the right of the vertical bar, once covariantly and once contravariantly (where occurrences to the left of the turnstile $\vdash$ are counted with opposite variance).

This notation is especially valuable when the same cat appears multiple times. For instance, if $M \in \mathcal{Mod}_{\mathbb{K}}(A, A^\circ, A)$ and $N \in \mathcal{Mod}_{\mathbb{K}}(A)$, then there are two kinds of extraordinary morphisms from M to N , which are represented in this notation as

$$(x : A), (y : A) \mid M(x, x, y) \vdash N(y)$$

$$(x : A), (y : A) \mid M(x, y, y) \vdash N(x)$$

It is also especially useful when talking about multivariable extraordinary morphisms. For instance, we could have

$$(a : A), (b : B), (c : C), (d : D) \mid M_1(a, b), M_2(b, c) \vdash N(a, d, c, d)$$

but here there is no way to act on M_1 and M_2 by symmetric groups separately to put the B, B° in front. Formally, this means that when describing multivariable extraordinary morphisms we have to incorporate a permutation in the definition; but with abstract variable notation we don't need to think about that.

Finally, if the cats that the variables correspond to are obvious from context, we can omit their declarations, obtaining a more concise notation $M(a, b, a) \vdash N(b, c, c)$ for a set of extraordinary module morphisms. Thus, if we want to give a name to a particular such morphism we can write

$$\phi \in (M(a, b, a) \vdash N(b, c, c))$$

As noted in ??, the symbol $\vdash$ is still present here to indicate the use of abstract variables, even though the correspondence of variables to cats is not explicitly “declared”.

It is worth emphasizing one difference between the use of abstract variables for restriction and for extraordinary morphisms. In the case of restriction, we write $(a : A), (b : B) \vdash M(f(a), g(b))$ to denote the *actual module* $(f, g)^*M$ obtained by restricting M along f and g . However, in the case of extraordinary morphisms, it is probably better not to think of the notation $(a : A) \vdash M(a, a)$ as actually denoting any module distinct from M ;¹ in particular it is not obtained by restriction of M along any sort of “diagonal” (which need not exist in an arbitrary kit). We will also combine these notations, writing for instance $M(f(a), g(a), h(b)) \vdash N(k(b))$; in this case it is better to think of the restrictions as happening first (obtaining modules such as $(a_1 : A), (a_2 : A), (b : B) \vdash M(f(a_1), g(a_2), h(b))$), followed by the pairing of variables to determine a type of extraordinary morphism.

So far we have described our “abstract variable notation” somewhat informally. When doing formal category theory, this informal understanding is generally sufficient. However, the notation can in fact be made much more precise, by defining a type theory to serve as the “internal language” of a kit; see [?].

4. COMPOSING EXTRAORDINARY MORPHISMS

The simplest sort of composition for extraordinary morphisms is composition with ordinary ones on either side. For instance, a morphism $\phi \in (M(a, b, a) \vdash N(b, c, c))$ can be composed with an ordinary map $\psi : N \rightarrow N'$ in $\mathcal{Mod}_{\mathbb{K}}(B, C, C^\circ)$ to obtain

¹In the formal definition to be presented in ??, it is possible to think of $(a : A) \vdash M(a, a)$ as an object of “ $\mathcal{SE}$ ”, but this is arguably just a technical device.

a morphism $\psi\phi \in (M(a, b, a) \vdash N'(b, c, c))$ In examples it is easy to see how to do this: just compose the components

$$M(a, b, a) \xrightarrow{\phi_{a,b,c}} N(b, c, c) \xrightarrow{\psi_{b,c,c}} N'(b, c, c)$$

and check that the naturality relations (both ordinary and extraordinary) are preserved. (Note that the components ψ_{b,c_1,c_2} with $c_1 \neq c_2$ do not appear in the definition, but they do appear in the proof of extraordinary naturality.) Formally, we can say that each set of extraordinary morphisms admits an action on both sides by the categories $\mathcal{Mod}_{\mathbb{K}}(\Psi)$ (that is, it is a profunctor).

More interesting sorts of composition involve two extraordinary transformations, as explained in [?]. The most basic composite of this sort involves two morphisms

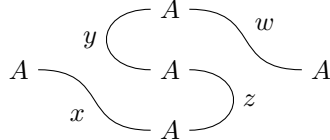
$$\phi \in (M(x) \vdash N(x, y, y)) \quad \psi \in (N(z, z, w) \vdash P(w)).$$

with x, y, z, w all associated to the same cat A . In examples, we can compose these by composing their components

$$M(x) \xrightarrow{\phi_{x,x}} N(x, x, x) \xrightarrow{\psi_{x,x}} P(x)$$

and check that the result is an ordinary natural transformation $M \rightarrow P$.

Note that the identification of all three variables in N is mandated in order to be able to write both ϕ and ψ , since ϕ requires the second and third variables to be the same while ψ requires the first and second to be the same. This then determines how the naturality of the composite matches up the variables of M and P (in this case, of course, they have only one variable each, but in general they might have many and this matching might be ambiguous). This process of identifying variables can be represented by drawing a graph:



or by solving a system of equations that “unifies” each arguments of the codomain $N(x, y, y)$ of ϕ with the corresponding argument of the domain $N(z, z, w)$ of ψ :

$$x = z \quad y = z \quad y = w$$

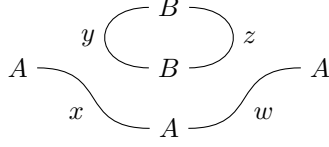
There are cases where this doesn’t work; that is, pairs of extraordinary morphisms from M to N and from N to P that are “non-composable” despite having the same module N in the middle. For example, we might instead have

$$\phi \in (M(x) \vdash N(x, y, y)) \quad \xi \in (N(w, z, z) \vdash P(w)).$$

The problem here is that when “composing components” there is no way to decide what y and z should be: we can write down the composite

$$M(x) \xrightarrow{\phi_{x,y,y}} N(x, y, y) \xrightarrow{\xi_{x,y,y}} P(x)$$

for any value of y , and there is no canonical choice (note that x and w might belong to an entirely different cat, so choosing $y = x$ is not an option). Graphically, this problem is visible as the presence of a “loop”:



A more careful explanation of this condition, and a proof that it is the most general answer (for the classical case of enriched categories), can be found in [?].

The technical work in the precise definition of a kit is to make precise *all* the allowable composites of extraordinary morphisms, how they interact with restriction along functors, and in what sense these composites must be “associative”. However, in practical applications this generality is rarely needed, as most extraordinary morphisms only involve a few variables, and intuition from ordinary category theory is generally sufficient to give the right answer. We intend to demonstrate this with some examples of doing “formal category theory” in a kit; but first we need to discuss some universal properties of cats and modules.

5. UNIVERSAL CONSTRUCTIONS

Universal constructions in a kit are more complicated than those in an ordinary category, due to the interaction of the different kinds of morphisms. Specifically, universal properties of cats generally have to be extended to say something about modules and module morphisms, while universal properties of modules have to include the extraordinary module morphisms somehow and also be stable under restriction along functors.

5.1. Tensor products of cats and opposites. We begin with the simplest universal properties for cats: those that directly internalize the operations represented “virtually” by the multicategorical nature of functors. For instance, a tensor product of cats A, B should be a cat $A \otimes B$ with a functor $\chi : (A, B) \rightarrow A \otimes B$ through which any functor $(A, B) \rightarrow C$ factors uniquely.

However, this plain universal property, though it suffices to determine $A \otimes B$ uniquely up to isomorphism, is insufficient in several ways. Firstly, as in the case of ordinary multicategories, it needs to apply to more general functors containing (A, B) as only part of their domain. Secondly, it needs to apply even in the contravariant case: functors $(A^\circ, B^\circ) \rightarrow C$ should factor through $(A \otimes B)^\circ$. Thirdly, it needs to apply to modules as well: we should have $\text{Mod}_{\mathbb{K}}(A \otimes B) \cong \text{Mod}_{\mathbb{K}}(A, B)$, and similarly when other cats are also present. And finally, it also needs to apply to module morphisms of all sorts, even extraordinary ones.

In order to state all of these universal properties precisely, we need some more notation.

- We will write A^ε to indicate either A or A° , and in this case we write A^{ε^*} for the other one (A° or A , respectively).
- We will use the letter Γ for a finite list of modules with abstract variables, such as $M(a, b), N(b, c, d), P(e)$ or $M_1(f(x), g(y, z)), M_2(h(y, x))$. Similarly, we will use the letter Ω for a single module with abstract variables; thus a general extraordinary module morphism can be written $\phi \in (\Gamma \vdash \Omega)$.
- We will write $\Gamma[x \leftarrow \theta]$ for the result of substituting the expression θ for all occurrences of the abstract variable x in Γ , and similarly, for Ω . Thus, for instance, if $\Omega = M(a, b)$, then $\Omega[a \leftarrow f(x, y)] = M(f(x, y), b)$. Note that this

might perform zero, one, or two substitutions, depending on how many times the variable occurs.

Definition 5.1. A **tensor product** of cats A, B is a cat $A \otimes B$ with a functor $\chi : (A, B) \rightarrow A \otimes B$ such that

- (i) Precomposition with χ induces bijections of functor sets

$$\mathbb{K}((\Psi, (A \otimes B)^\varepsilon), C) \simeq \mathbb{K}((\Psi, A^\varepsilon, B^\varepsilon), C)$$

for any list Ψ of cats with variance;

- (ii) Restriction along χ induces an isomorphism of categories

$$\mathcal{Mod}_{\mathbb{K}}(\Psi, (A \otimes B)^\varepsilon) \simeq \mathcal{Mod}_{\mathbb{K}}(\Psi, A^\varepsilon, B^\varepsilon)$$

for any Ψ ; and finally

- (iii) Restriction along χ induces bijections of sets of extraordinary morphisms

$$(\Psi, z : (A \otimes B)^\varepsilon \mid \Gamma \vdash \Omega) \simeq (\Psi, x : A^\varepsilon, y : B^\varepsilon \mid \Gamma[z \leftarrow \chi(x, y)] \vdash \Omega[z \leftarrow \chi(x, y)])$$

whenever this makes sense.

Note that due to the symmetric group actions, it suffices to assert these universal properties when $A \otimes B$ is at the end of a list; corresponding properties then follow when it appears anywhere.

We can similarly define n -ary tensor products of cats for all n , and using only ?? we can show that these are associative and symmetric insofar as they exist. That is, if $A \otimes B$ and $(A \otimes B) \otimes C$ exist, then the latter is a ternary tensor product; hence if $B \otimes C$ exists then $A \otimes (B \otimes C)$ exists and is isomorphic to $(A \otimes B) \otimes C$. A 0-ary tensor product is a cat I with a functor $\chi : () \rightarrow I$, precomposition with which induces isomorphisms

$$\begin{aligned} \mathbb{K}((\Psi, I^\varepsilon), C) &\simeq \mathbb{K}(\Psi, C) \\ \mathcal{Mod}_{\mathbb{K}}(\Psi, I^\varepsilon) &\simeq \mathcal{Mod}_{\mathbb{K}}(\Psi) \\ (\Psi, z : I^\varepsilon \mid \Gamma \vdash \Omega) &\simeq (\Psi \mid \Gamma[z \leftarrow \chi] \vdash \Omega[z \leftarrow \chi]) \end{aligned}$$

In particular, if all tensor products of cats exist, then the category $\mathbb{K}_1$ of cats and unary functors is symmetric monoidal.

Similarly, we can internalize contravariance by defining an operation of “oppositization”. We will write this as $(-)^{\text{op}}$ to distinguish it from the formal marker of contravariance $(-)^\circ$.

Definition 5.2. An **opposite** of a cat A is a cat A^{op} with a functor $\chi : (A^\circ) \rightarrow A^{\text{op}}$ precomposition with which induces isomorphisms

$$\begin{aligned} \mathbb{K}((\Psi, (A^{\text{op}})^\varepsilon), C) &\simeq \mathbb{K}((\Psi, A^{\varepsilon^*}), C) \\ \mathcal{Mod}_{\mathbb{K}}(\Psi, (A^{\text{op}})^\varepsilon) &\simeq \mathcal{Mod}_{\mathbb{K}}(\Psi, A^{\varepsilon^*}) \\ (\Psi, z : (A^{\text{op}})^\varepsilon \mid \Gamma \vdash \Omega) &\simeq (\Psi, x : A^{\varepsilon^*} \mid \Gamma[z \leftarrow \chi(x)] \vdash \Omega[z \leftarrow \chi(x)]) \end{aligned}$$

If all opposites exist, then $(-)^{\text{op}}$ defines an endofunctor of $\mathbb{K}_1$. By comparing universal properties, we can see that $(A^{\text{op}})^{\text{op}} \cong A$ and $(A \otimes B)^{\text{op}} \cong A^{\text{op}} \otimes B^{\text{op}}$ and I^{op} , so that $(-)^{\text{op}}$ is a “symmetric monoidal involution” in a suitable sense.

5.2. Tensor products of modules. Like the tensor product of cats, the tensor product of modules has a universal property very much like that of tensor products in an ordinary multicategory. It is simpler in that we don't need to worry about the dependence of modules on cats (the second two properties in ???). However, it is more complicated in that the universal property applies to all sorts of extraordinary morphisms, and that it must be stable under restriction along functors.

To state this precisely, we need a bit more notation.

- If $M \in \mathcal{Mod}_{\mathbb{K}}(\Psi)$, we will write $M(\theta)$ for M with a list of expressions θ for its abstract variables. For instance, if $M \in \mathcal{Mod}_{\mathbb{K}}(A, B^\circ, C)$ and we have $g : X \rightarrow A$ and $h : (Y^\circ, Z) \rightarrow B$ and $k : () \rightarrow C$, then with $\theta = (g(x), h(y, z), k)$ we have $M(\theta) = M(g(x), h(y, z), k)$. We will also use letters like γ and ω for such lists θ .

Definition 5.3. A **tensor product** of modules $M_1 \in \mathcal{Mod}_{\mathbb{K}}(\Psi_1)$ and $M_2 \in \mathcal{Mod}_{\mathbb{K}}(\Psi_2)$ is a module $M_1 \otimes M_2 \in \mathcal{Mod}_{\mathbb{K}}(\Psi_1, \Psi_2)$ along with an (ordinary) morphism $\chi : (M_1, M_2) \rightarrow M_1 \otimes M_2$ such that precomposition with χ (or some restriction of it) induces bijections

$$(\Gamma, (M_1 \otimes M_2)(\theta_1, \theta_2) \vdash \Omega) \xrightarrow{\sim} (\Gamma, M_1(\theta_1), M_2(\theta_2) \vdash \Omega)$$

The presence of the θ s means that this universal property includes extraordinary morphisms of all sorts, where paired variables may occur anywhere. Examples include:

$$\begin{aligned} ((M_1 \otimes M_2)(a, b) \vdash N(a, b)) &\xrightarrow{\sim} (M_1(a), M_2(b) \vdash N(a, b)) \\ ((M_1 \otimes M_2)(a, a) \vdash N) &\xrightarrow{\sim} (M_1(a), M_2(a) \vdash N) \\ (P(a), (M_1 \otimes M_2)(a, b, c) \vdash N(b, c, d, d)) &\xrightarrow{\sim} (P(a), M_1(a, b), M_2(c) \vdash N(b, c, d, d)) \end{aligned}$$

and so on. Similarly, θ can also include restrictions, so that we might have

$$((M_1 \otimes M_2)(f(a), g(b)) \vdash N(a, b)) \xrightarrow{\sim} (M_1(f(a)), M_2(g(b)) \vdash N(a, b))$$

In other words, the restricted morphism $(f, g)^*\chi : f^*M, g^*N \rightarrow (f, g)^*(M \otimes N)$ has the universal property that would be expected of $\chi : f^*M, g^*N \rightarrow f^*M \otimes g^*N$, and thus tensor products are stable under restriction: $(f, g)^*(M \otimes N) \cong f^*M \otimes g^*N$.

In general this stability is only up to isomorphism, so we technically ought to resist the temptation to write things like $M_1(a, b) \otimes M_2(c)$ instead of $(M_1 \otimes M_2)(a, b, c)$. However, the former notation is so useful (it reminds us how the dependencies split) that we will often succumb to this temptation. It should be possible to prove a coherence theorem making tensor products (and all similar universal properties to be introduced below) strictly stable under restriction, and this is moreover how they appear in the type theory of [?]. Remember also that the pairing of variables only indicates the type of extraordinary morphism under consideration; when we write $(M_1 \otimes M_2)(a, a)$ or $M_1(a) \otimes M_2(a)$ we are still talking about the same tensor product module $M_1 \otimes M_2$, just different kinds of morphisms out of it.

As usual, we can define n -ary tensor products for all n , and the usual arguments show that these are associative and symmetric insofar as they exist. A nullary tensor product is a module $I \in \mathcal{Mod}_{\mathbb{K}}()$ and a morphism $() \rightarrow I$, precomposition with which induces bijections $(\Gamma, I \vdash \Omega) \xrightarrow{\sim} (\Gamma \vdash \Omega)$.

5.3. Ends and coends. The tensor product of modules has a universal property that extends to extraordinary morphisms, but the universal morphism $(M, N) \rightarrow M \otimes N$ itself is an ordinary morphism. Now we consider universal morphisms that are themselves extraordinary. Perhaps the simplest of these are ends and coends. First one more bit of notation:

- We will write ξ for a θ that contains no functors or pairings, only a list of distinct abstract variables. In other words, $M(\xi)$ simply stands for a way to import M into abstract variable notation, which is unique up to the choice of variable names. Thus if $M \in \mathcal{Mod}_{\mathbb{K}}(A, B^\circ, C)$ then $M(\xi)$ might mean $M(a, b, c)$ but not $M(g(x), h(y, z), k)$. Moreover, if different ξ s appear in an expression we assume them to be disjoint.

For instance, the universal morphism of a tensor product could be written in abstract variable notation as $\chi \in (M_1(\xi_1), M_2(\xi_2) \vdash (M_1 \otimes M_2)(\xi_1, \xi_2))$. We avoided this before by writing it without abstract variables, but that is no longer possible now that we want to consider extraordinary universal morphisms.

Definition 5.4. A **coend** of a module $M \in \mathcal{Mod}_{\mathbb{K}}(A, A^\circ, \Psi)$ is a module $\int^A M \in \mathcal{Mod}_{\mathbb{K}}(\Psi)$ together with a morphism $\chi \in (M(a, a, \xi) \vdash (\int^A M)(\xi))$, precomposition with which induces bijections

$$(\Gamma, (\int^A M)(\theta) \vdash \Omega) \xrightarrow{\sim} (\Gamma, M(a, a, \theta) \vdash \Omega)$$

whenever this makes sense.

If other variables belonging to A occur elsewhere, then we may indicate which one is being “coended” by writing $\int^{a:A} M(a, a)$. In this notation the variable a is “bound” and no longer indicates any ordinary or extraordinary dependence of the morphism under consideration.

The simplest case of the universal property of a coend is that

$$(\int^A M \vdash N) \cong (M(a, a) \vdash N)$$

for $M \in \mathcal{Mod}_{\mathbb{K}}(A, A^\circ)$ and $N \in \mathcal{Mod}_{\mathbb{K}}()$. In other words, $\int^A M$ is the module universally equipped with a morphism from M that is extraordinary in A . As with tensor products, the full universal property also applies to multivariable morphisms that may have other variables, both ordinary and extraordinary, and also implies a stability under restriction: $f^*(\int^A M) \cong \int^A f^*M$.

This stability is, of course, only true for restriction along functors between *other* cats that M may depend on: a functor $f : B \rightarrow A$ does not induce an isomorphism $\int^B (f, f)^*M \cong \int^A M$. We do, however, have a morphism $\int^B (f, f)^*M \rightarrow \int^A M$, induced by first restricting the universal morphism $M(a, a) \rightarrow \int^A M$ along f to obtain $M(f(b), f(b)) \rightarrow \int^A M$, then applying the universal property of $\int^B$.

By comparing universal properties, we can deduce the “Fubini theorem”

$$\int^A \int^B M \cong \int^B \int^A M.$$

Moreover, if $A \otimes B$ exists, then both iterated coends are also isomorphic to $\int^{A \otimes B} M'$, where M' corresponds to M under the universal property of $A \otimes B$ (applied twice, once covariantly and once contravariantly). Similarly, we have $\int^I M' \cong M$ if I is the unit cat and M' corresponds to M under its universal property.

Combining the coend with the (external) tensor product of modules, we obtain the *binding* or *canceling* tensor product of modules:

$$M \otimes_A N = \int^A (M \otimes N)$$

where $M \in \mathcal{Mod}_{\mathbb{K}}(\Psi_1, A)$ and $N \in \mathcal{Mod}_{\mathbb{K}}(\Psi_2, A^\circ)$ (or vice versa). Comparing universal properties, we see that this is also associative:

$$(M \otimes_A N) \otimes_B P \cong M \otimes_A (N \otimes_B P)$$

and we can verify the pentagon axiom. This wants to be the composition operation in a bicategory of cats and modules, where the hom-category from A to B is $\mathcal{Mod}_{\mathbb{K}}(B^\circ, A)$; but we do not yet have the identity 1-cells for such a bicategory.

Note that it is possible to express directly the universal property of the binding tensor product: it comes with a morphism $M(\xi_1, a), N(a, \xi_2) \vdash (M \otimes_A N)(\xi_1, \xi_2)$, inducing bijections

$$(\Gamma, (M \otimes_A N)(\theta_1, \theta_2) \vdash \Omega) \xrightarrow{\sim} (\Gamma, M(\theta_1, a), N(a, \theta_2) \vdash \Omega)$$

This is useful because in some kits the binding tensor product may exist even if the external tensor product and coend do not separately exist.

Finally, ends are dual to coends. These are our first module with a “mapping in” universal property.

Definition 5.5. An **end** of a module $M \in \mathcal{Mod}_{\mathbb{K}}(A, A^\circ, \Psi)$ is a module $\int_A M \in \mathcal{Mod}_{\mathbb{K}}(\Psi)$ together with a morphism $\chi \in ((\int_A M)(\xi) \vdash M(a, a, \xi))$, postcomposition with which induces bijections

$$(\Gamma \vdash (\int_A M)(\theta)) \xrightarrow{\sim} (\Gamma \vdash M(a, a, \theta))$$

whenever this makes sense.

5.4. Function modules. For ends and coends, the universal morphism is extraordinary in the input (the module being ended or coended); now we move on to universal morphisms that are extraordinary in the output (the object with a universal property). The first of these are function modules, the “right adjoints” to tensor products — although as we will see, to actually construct adjunctions requires incorporating ends and coends as well.

Definition 5.6. A **function module** of modules $M_1 \in \mathcal{Mod}_{\mathbb{K}}(\Phi_1)$ and $M_2 \in \mathcal{Mod}_{\mathbb{K}}(\Phi_2)$ is a module $M_1 \multimap M_2 \in \mathcal{Mod}_{\mathbb{K}}(\Phi_1^\circ, \Phi_2)$ with a morphism

$$\text{ev} \in ((M_1 \multimap M_2)(\xi_1, \xi_2), M_1(\xi_1) \vdash M_2(\xi_2)),$$

postcomposition with which induces bijections

$$(5.7) \quad (\Gamma \vdash (M_1 \multimap M_2)(\theta_1, \theta_2)) \xrightarrow{\sim} (\Gamma, M_1(\theta_1) \vdash M_2(\theta_2))$$

whenever this makes sense.

As before, ξ ’s represent disjoint lists of distinct variables; thus ev (“evaluation”) is a single morphism such as $(M_1 \multimap M_2)(a, b), M_1(a) \vdash M_2(b)$. But also as before, the θ ’s represent arbitrary expressions, possibly with variable pairings; thus the

universal property includes as particular cases:

$$\begin{aligned}
(N(a, b) \vdash (M_1 \multimap M_2)(a, b)) &\xrightarrow{\sim} (N(a, b), M_1(a) \vdash M_2(b)) \\
(N \vdash (M_1 \multimap M_2)(a, a)) &\xrightarrow{\sim} (N, M_1(a) \vdash M_2(a)) \\
(N(a, b) \vdash (M_1 \multimap M_2)(a, c, b, c)) &\xrightarrow{\sim} (N(a, b), M_1(a, c) \vdash M_2(b, c)) \\
(N(f(x, z), g(y)) \vdash (M_1 \multimap M_2)(h(x, y), z)) &\xrightarrow{\sim} (N(f(x, z), g(y)), M_1(h(x, y)) \vdash M_2(z))
\end{aligned}$$

and so on. Note that the composition map (??) is indeed what we get by postcomposing with ev : as described in ??, when composing $((M_1 \multimap M_2)(\xi_1, \xi_2), M_1(\xi_1) \vdash M_2(\xi_2))$ with $(\Gamma \vdash (M_1 \multimap M_2)(\theta_1, \theta_2))$ (and technically the identity on $M_1(\xi_1)$ as well), ξ_1 and ξ_2 get identified with θ_1 and θ_2 , so that in the result instead of $M_1(\xi_1)$ and $M_2(\xi_2)$ in the domain and codomain we have $M_1(\theta_1)$ and $M_2(\theta_2)$.

Combining the universal properties of function modules with tensor products, we have an isomorphism

$$((M_1 \otimes M_2)(\theta_1, \theta_2) \vdash M_3(\theta_3)) \cong (M_1(\theta_1) \vdash (M_2 \multimap M_3)(\theta_2, \theta_3))$$

that looks like an adjunction, except that one side or the other (or both) must always be extraordinary. However, if we add in an end or a coend, we can make an isomorphism between two ordinary sets of morphisms, and therefore an actual adjunction. Examples include

$$\begin{aligned}
\left(\int^{a:A} (M_1 \otimes M_2)(a, a) \vdash M_3 \right) &\cong (M_1(a) \vdash (M_2 \multimap M_3)(a)) \\
((M_1 \otimes M_2)(a) \vdash M_3(a)) &\cong (M_1 \vdash \int_{a:A} (M_2 \multimap M_3)(a, a))
\end{aligned}$$

which can be interpreted (for fixed $M_2 \in \text{Mod}_{\mathbb{K}}(A)$) as adjunctions

$$\begin{aligned}
\text{Mod}_{\mathbb{K}}(A) &\rightleftarrows \text{Mod}_{\mathbb{K}}(()) \\
\text{Mod}_{\mathbb{K}}(()) &\rightleftarrows \text{Mod}_{\mathbb{K}}(A)
\end{aligned}$$

respectively. Adding in a couple of other variables, we find that for $M \in \text{Mod}_{\mathbb{K}}(B^\circ, A)$ and $N \in \text{Mod}_{\mathbb{K}}(C^\circ, B)$ and $P \in \text{Mod}_{\mathbb{K}}(C^\circ, A)$ we have

$$\left(\int^{b:B} (N \otimes M)(c, b, b, a) \vdash P(c, a) \right) \cong (M(b, a) \vdash \int_{c:C} (N \multimap P)(c, b, c, a))$$

Since $\int^B (N \otimes M)$ is the composition in the expected “bicategory of profunctors”, this shows that if we have function modules and ends then that will be a *closed bicategory*, in the usual sense that its composition functors have right adjoints on both sides.

As in the case of the binding tensor product, it is sometimes useful to directly name the *binding function module*

$$M \multimap_A N = \int_A (M \multimap N)$$

whose universal property can be expressed directly as a morphism

$$\text{ev} \in ((M \multimap_A N)(\theta_1, \theta_2), M(\theta_1, a) \vdash N(\theta_2, a))$$

postcomposition with which induces an isomorphism

$$(\Gamma \vdash (M \multimap_A N)(\theta_1, \theta_2)) \xrightarrow{\sim} (\Gamma, M(\theta_1, a) \vdash N(\theta_2, a)).$$

5.5. Hom-modules and representable modules. The most important universal objects in a kit are the “hom-modules” $\text{hom}_A \in \mathcal{M}od_{\mathbb{K}}(A^\circ, A)$. Their universal property is essentially a “two-sided double Yoneda lemma”.

Definition 5.8. A **hom-module** of a cat A is a module $\text{hom}_A \in \mathcal{M}od_{\mathbb{K}}(A^\circ, A)$ together with a morphism $\text{id}_A \in (\vdash \text{hom}_A(a, a))$, precomposition with which induces bijections

$$(\Gamma, \text{hom}_A(x, y) \vdash \Omega) \xrightarrow{\sim} (\Gamma[y \leftarrow x] \vdash \Omega[y \leftarrow x])$$

Two instances of this universal property look like the Yoneda lemma and the co-Yoneda lemma:

$$\begin{aligned} (\text{hom}_A(x, y) \vdash M(x, y)) &\xrightarrow{\sim} (\vdash M(x, x)) \\ (M(x), \text{hom}_A(x, y) \vdash N(y)) &\xrightarrow{\sim} (M(x) \vdash N(x)) \end{aligned}$$

TODO: Derive the one-sided versions.

5.6. Functor and presheaf cats.

6. EXAMPLES OF FORMAL CATEGORY THEORY

We will always assume that all kits have hom-modules; this seems to be a prerequisite for doing much of any formal category theory. However, we will explicitly assume other universal objects when we need them.

6.1. Limits and colimits. Consider functors $d : D \rightarrow C$ and $l : () \rightarrow C$ (i.e. l is an “object” of C), and a module $J \in \mathcal{M}od_{\mathbb{K}}(D)$. A **J -weighted cone** over d with vertex l is a (ordinary) module morphism

$$\pi \in (J(x) \vdash \text{hom}_C(l, d(x))) .$$

Using the universal property of hom-modules, this corresponds to a morphism

$$\pi' \in (\text{hom}_C(y, l), J(x) \vdash \text{hom}_C(y, d(x))) .$$

We say that π exhibits l as a **J -weighted limit** of d if π' gives $\text{hom}_C(1, l)$ the universal property of the binding function module $J \multimap_D \text{hom}_C(1, d)$; in other words, if postcomposition with π' induces bijections

$$(\Gamma \vdash \text{hom}_C(y, l)) \xrightarrow{\sim} (\Gamma, J(x) \vdash \text{hom}_C(y, d(x))) .$$

Of course, if $J \multimap_D \text{hom}_C(1, d)$ is known to exist (such as if $\mathbb{K}$ has ends and function modules), then to make l a J -weighted limit of d is equivalent to giving an isomorphism $\text{hom}_C(1, l) \cong J \multimap_D \text{hom}_C(1, d)$ in $\mathcal{M}od_{\mathbb{K}}(C^\circ)$.

(Hmm, in the definition of function modules (and all the other universal properties) I incorporated a substitution to make it restriction-stable. Is that necessary to do here too? In an equipment that corresponds to asserting the universal property with respect to cells having nontrivial vertical-arrow boundaries. But maybe we can avoid that once we have homs and the Yoneda lemma by moving them into the domain?)

While the above definition most closely matches the usual definition of weighted limits, it is not necessary to restrict d and l to have unary and nullary domains,

respectively. The case when l also has unary domain E is also very important. In this case π is ordinary in E and π' is extraordinary:

$$\begin{aligned}\pi &\in (J(e, x) \vdash \text{hom}_C(l(e), d(x))) \\ \pi' &\in (\text{hom}_C(y, l(e)), J(e, x) \vdash \text{hom}_C(y, d(x)))\end{aligned}$$

and the limit is again characterized by $\text{hom}_C(1, l) \cong J \multimap_D \text{hom}_C(1, d)$, only now this is an isomorphism in $\text{Mod}_{\mathbb{K}}(C^\circ, E)$. Since universal properties are stable under restriction, it follows that for any $f : E' \rightarrow E$ we have also $\text{hom}_C(1, lf) \cong J(f, 1) \multimap_D \text{hom}_C(1, d)$, so that $l : E \rightarrow C$ is a functor given by taking “pointwise limits”.

As a special case, we can define (pointwise) Kan extensions. If $f : D \rightarrow C$ and $g : D \rightarrow E$ are functors, then the right Kan extension of f along g (if it exists) is the limit of f weighted by $\text{hom}_E(1, g)$.

Part 2. The definition of a kit

7. GENERALIZED POLYCATEGORIES

Our formal definition of a kit will be a special case of a general notion called a *generalized polycategory*. Generalized polycategories appear only implicitly in the literature (e.g. [?]), so we will have to define them as well along the way. However, we defer their general study to a later paper, taking here a reasonably geodesic route to the definition of kit.

Roughly speaking, generalized polycategories are to polycategories as generalized multicategories are to multicategories. Recall that a *multicategory* is like a category in which a morphism can have a finite list of objects, rather than a single object, as its domain (but still only a single object as codomain). *Generalized multicategories* are a vast generalization obtained by observing that “finite lists” can be regarded as elements of the free-monoid monad, and that a version of the definition of multicategory can be carried out for *any* sufficiently well-behaved monad.

Polycategories, on the other hand, are like categories in which a morphism can have finite lists of objects as both its domain and codomain. However, this brief description is significantly farther from a precise definition than the corresponding brief description of multicategories, because there are many more choices about how to allow such morphisms to be composed. Polycategories proper [?, ?] result from one such choice, but other (perhaps even more natural) choices yield structures such as “colored props” and “colored properads”.²

Generalized polycategories are thus defined relative not just to two monads (one describing the domains and one the codomains), but also a “distributive law” between them that describes “how morphisms should be composed”. Ordinary polycategories were defined in this language by Garner [?]; to obtain a general notion we abstract out the details of his construction (with some changes; see below). This is promising as an avenue for defining kits, since we can use the distributive law to encode the “allowable composites” of extraordinary natural transformations.

There are a number of perspectives on generalized multicategories that one could enhance to a theory of generalized polycategories. We will use the framework of [?],

²Thus perhaps the general notion should be called a “generalized prop”, but “generalized polycategory” is more euphonious and intuitive-sounding.

except that (aiming at a geodesic route to kits) we simplify it by considering only double categories rather than the “virtual double categories” used there.

Specifically, we consider *pseudo double categories*, in which one direction of composition is strict and the other is only associative and unital up to isomorphisms in the strict direction. The literature is not consistent about whether the strict direction is the “vertical” or the “horizontal” one. To be agnostic about this, we will use instead the words **tight** and **loose** (from [?]) for the strict and pseudo directions, and we allow ourselves to write them in either orientation, marking the loose morphisms with a dot to disambiguate. Thus, a cell in a double category can be written either as

$$\begin{array}{ccc} A & \xrightarrow{\bullet} & B \\ \downarrow & \Downarrow & \downarrow \\ C & \xrightarrow{\bullet} & D \end{array} \quad \text{or} \quad \begin{array}{ccc} A & \longrightarrow & C \\ \downarrow \bullet & \Rightarrow & \downarrow \bullet \\ B & \longrightarrow & D \end{array}$$

We write tight composition (of arrows or cells) with $\cdot$ or juxtaposition in applicative order, and loose composition (of arrows or cells) with $\bullet$ in diagrammatic order.³ Similarly, we write 1 for tight identities (of objects or loose arrows), and Id_A for loose identities (of objects or tight arrows).

The particular double category we have most in mind is $\mathbb{Cat}$, where:

- The objects are categories,
- The tight arrows are functors,
- A loose arrow $A \dashrightarrow B$ is a profunctor, i.e. a functor $B^{\text{op}} \times A \rightarrow \mathbf{Set}$, and
- The 2-cells are natural transformations of the obvious sort.

We will also be interested in *slice* double categories of $\mathbb{Cat}$: for any object A of a double category $\mathbb{D}$, in the slice $\mathbb{D}/A$ the category $(\mathbb{D}/A)_0$ of objects and tight maps is the slice of $\mathbb{D}_0$ over A , and the category $(\mathbb{D}/A)_1$ of loose arrows and cells is the slice of $\mathbb{D}_1$ over Id_A .

Between (pseudo) double categories we will mainly be interested in *strong* or *pseudo double functors*, which are strictly functorial in the tight direction and functorial up to tight isomorphism in the loose direction. Such functors admit an obvious notion of *tight transformation*, with tight arrow components $\alpha_A : FA \rightarrow GA$ for each object A , and 2-cell components

$$\begin{array}{ccc} FA & \xrightarrow{FM} & FB \\ \alpha_A \downarrow & \Downarrow & \downarrow \alpha_B \\ GA & \xrightarrow{GM} & GB \end{array}$$

for each loose arrow $A \dashrightarrow B$. This yields a strict 2-category of double categories which we denote $\mathcal{Dbl}$. By a **double monad** we mean a monad in this 2-category. (Many of the definitions make sense even for double monads whose underlying double functors are *lax*, but all the ones we will be interested in are strong.)

Distributive laws can also be defined internal to any 2-category; interpreted in $\mathcal{Dbl}$ this yields a notion of **tight distributive law**. However, for generalized poly-categories we are interested in *loose distributive laws*, which we now define. If F and

³Applicative order means that the composite of $f : A \rightarrow B$ and $g : B \rightarrow C$ is $g \cdot f$, while diagrammatic order means that the composite of $M : A \dashrightarrow B$ and $N : B \dashrightarrow C$ is $M \bullet N$.

G are double functors between pseudo double categories, a **loose transformation** $\alpha : F \rightarrowtail G$ consists of a loose arrow $\alpha_X : FX \rightarrowtail GX$ for each object X of the domain, forming a pseudonatural transformation between the loose parts of F and G , together with for each tight arrow $f : X \rightarrow Y$ in the domain, a cell

$$\begin{array}{ccc} FX & \xrightarrow{\alpha_X} & GX \\ Ff \downarrow & \alpha_f & \downarrow Gf \\ FY & \xrightarrow{\alpha_Y} & GY \end{array}$$

satisfying some evident axioms; see [?] wherein this is called a *vertical transformation*. These are the loose arrows in a double category of functors whose tight arrows are tight transformations, and whose cells are an evident kind of “square modification”.

In the following definitions, we write composition of double functors (and whiskering of tight and loose transformations by double functors) with $\circ$, to distinguish it from the compositions $\cdot$ and $\bullet$ in the tight and loose direction respectively. A further reason for this will become apparent in ???. For brevity, we will also omit mention of the associativity and unit constraints for loose composition, and the functoriality constraints for double functors.

Definition 7.1. Let T and S be monads on a double category $\mathbb{K}$. A **loose distributive law** between (T, ν, ι) and (S, μ, η) consists of the following:

- (i) A loose transformation $\delta : T \circ S \rightarrowtail S \circ T$.
- (ii) Square modifications

$$\begin{array}{ccc} T \circ T \circ S & \xrightarrow{T \circ \delta} & T \circ S \circ T \xrightarrow{\delta \circ T} S \circ T \circ T \\ \nu \circ S \downarrow & \bar{\nu} & \downarrow S \circ \nu \\ T \circ S & \xrightarrow{\delta} & S \circ T \end{array} \quad \begin{array}{ccc} S & \xrightarrow{\text{Id}_S} & S \\ \iota \circ S \downarrow & \bar{\iota} & \downarrow S \circ \iota \\ T \circ S & \xrightarrow{\delta} & S \circ T \end{array}$$

$$\begin{array}{ccc} T \circ S \circ S & \xrightarrow{\delta \circ S} & S \circ T \circ S \xrightarrow{S \circ \delta} S \circ S \circ T \\ T \circ \mu \downarrow & \bar{\mu} & \downarrow \mu \circ T \\ T \circ S & \xrightarrow{\delta} & S \circ T \end{array} \quad \begin{array}{ccc} T & \xrightarrow{\text{Id}_T} & T \\ T \circ \eta \downarrow & \bar{\eta} & \downarrow \eta \circ T \\ T \circ S & \xrightarrow{\delta} & S \circ T \end{array}$$

These are versions of the diagrams that would have to *commute* if δ were an ordinary tight distributive law.

- (iii) The following equalities of square modifications hold:

$$\begin{array}{ll} \bar{\iota} \cdot \text{Id}_\eta = \bar{\eta} \cdot \text{Id}_\iota & \bar{\mu} \cdot ((\bar{\mu} \circ S) \bullet \mu_\delta) = \bar{\mu} \cdot (\delta_\mu \bullet (S \circ \bar{\mu})) \\ \bar{\mu} \cdot ((\bar{\eta} \circ S) \bullet \eta_\delta) = 1_\delta & \bar{\nu} \cdot (\nu_\delta \bullet (\bar{\nu} \circ T)) = \bar{\nu} \cdot ((T \circ \bar{\nu}) \bullet \delta_\nu) \\ \bar{\mu} \cdot (\delta_\eta \bullet (S \circ \bar{\eta})) = 1_\delta & \bar{\mu} \cdot ((\bar{\iota} \circ S) \bullet S \bar{\iota}) = \bar{\iota} \cdot \text{Id}_\mu \\ \bar{\nu} \cdot (\iota_\delta \bullet (\bar{\iota} \circ T)) = 1_\delta & \bar{\nu} \cdot ((T \circ \bar{\eta}) \bullet (\bar{\eta} \circ T)) = \bar{\eta} \cdot \text{Id}_\nu \\ \bar{\nu} \cdot ((T \circ \bar{\iota}) \bullet \delta_\iota) = 1_\delta & \bar{\mu} \cdot ((\bar{\nu} \circ S) \bullet (S \circ \bar{\nu})) = \bar{\nu} \cdot ((T \circ \bar{\mu}) \bullet (\bar{\mu} \circ T)) \cdot \bar{\delta} \end{array}$$

where

$$\bar{\delta} : (\delta \circ T \circ S) \bullet (T \circ S \circ \delta) \rightrightarrows (T \circ S \circ \delta) \bullet (\delta \circ T \circ S)$$

is a pseudonaturality constraint of δ . In all cases the tight naturality and monad axioms for the transformations ν, ι, μ, η ensures that the boundaries of both sides of the equations agree automatically.

The data and axioms of a loose distributive law may appear rather formidable when written out unforgivingly as above. In [??] we sketch them instead using an informal string diagram calculus, in which the monads T and S are written as a red and a blue string respectively and the loose transformation δ is a “string crossing”.

In [?] the theory of generalized multicategories relative to a double monad is developed in two steps, by first constructing a “horizontal Kleisli virtual double category” and then considering monoids therein. In a later paper we will do the same for generalized polycategories relative to a loose distributive law, but at present it suffices to define the end result explicitly.

In the following definitions we write application of double functors and transformations to objects by juxtaposition, so that for instance the components of $\delta : T \circ S \rightarrow S \circ T$ are of the form $\delta_A : TSA \rightarrow STA$.

Definition 7.2. Let $\delta : T \circ S \rightarrow S \circ T$ be a loose distributive law between double monads. A δ -**monoid** consists of an object A and a loose arrow $M : TA \rightarrow SA$ together with unit and multiplication cells

$$\begin{array}{ccc} A \xrightarrow{\text{Id}_A} A & TTA \xrightarrow{TM} TSA \xrightarrow{\delta} STA \xrightarrow{SM} SSA & \\ \downarrow \iota_A \quad e \Downarrow \quad \downarrow \eta_A & \downarrow \nu_A \quad m \Downarrow \quad \downarrow \mu_A & \\ TA \xrightarrow{M} SA & TA \xrightarrow{M} SA & \end{array}$$

such that the two following two ways to compose the multiplication with the unit are equal (modulo unit constraints for loose composition) to the identity on M :

$$\begin{array}{ccc} TA \xrightarrow{\text{Id}_{TA}} TA \xrightarrow{\text{Id}_{TA}} TA \xrightarrow{M} SA & TA \xrightarrow{M} SA \xrightarrow{\text{Id}_{SA}} SA \xrightarrow{\text{Id}_{SA}} SA & \\ \parallel \quad \bar{T} \quad \parallel \quad \parallel \quad \parallel & \parallel \quad \parallel \quad \parallel \quad \parallel \quad \bar{S} \quad \parallel & \\ TA \xrightarrow{T\text{Id}_A} TA \xrightarrow{\text{Id}_{TA}} TA \xrightarrow{M} SA & TA \xrightarrow{M} SA \xrightarrow{\text{Id}_{SA}} SA \xrightarrow{S\text{Id}_A} SA & \\ \downarrow T\iota_A \quad T\eta_A \quad \downarrow \bar{\eta} \quad \eta_{TA} \quad \eta_M \quad \downarrow \eta_{SA} \quad \downarrow \iota_A \quad \downarrow \iota_M \quad \downarrow \iota_{SA} \quad \downarrow \bar{\iota} \quad \downarrow S\iota_A \quad \downarrow S\eta_A & & \\ TTA \xrightarrow{TM} TSA \xrightarrow{\delta} STA \xrightarrow{SM} SSA & TTA \xrightarrow{TM} TSA \xrightarrow{\delta} STA \xrightarrow{SM} SSA & \\ \downarrow \nu_A \quad m \quad \downarrow \mu_A & \downarrow \nu_A \quad m \quad \downarrow \mu_A & \\ TA \xrightarrow{M} SA & TA \xrightarrow{M} SA & \end{array}$$

and also the two following ways to compose the multiplication with itself are equal to each other:

$$\begin{array}{ccc} TTTA \xrightarrow{TTM} TTSA \xrightarrow{T\delta} TSTA \xrightarrow{\delta T} STTA \xrightarrow{STM} STSA \xrightarrow{S\delta} SST A \xrightarrow{SSM} SSSA & & \\ \downarrow \nu_T \quad \downarrow \nu_M \quad \downarrow \nu_S \quad \downarrow \bar{\nu} \quad \downarrow S\nu \quad \downarrow S\mu & & \\ TTA \xrightarrow{TM} TSA \xrightarrow{\delta} STA \xrightarrow{SM} SSA & & \\ \downarrow \nu \quad m \quad \downarrow \mu & & \\ TA \xrightarrow{M} SA & & \end{array}$$

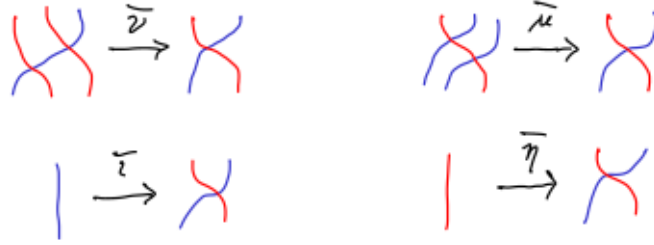


FIGURE 1. The data of a loose distributive law

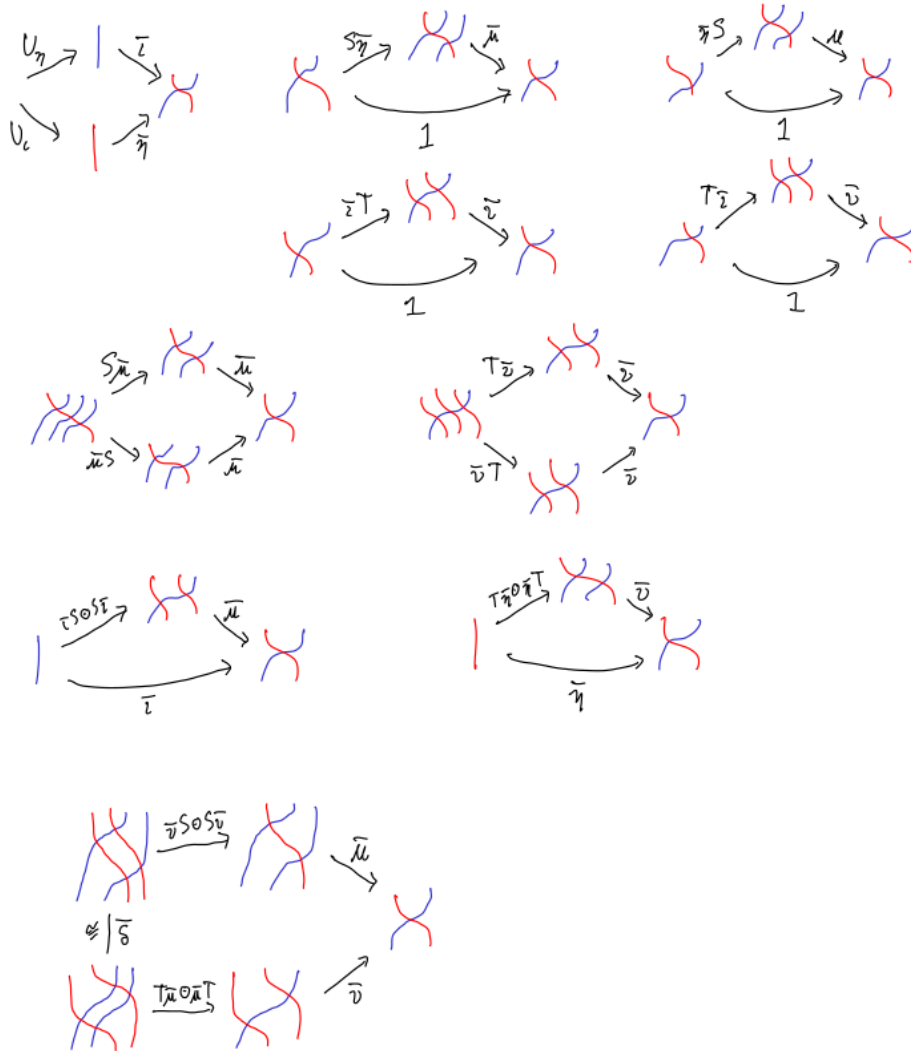


FIGURE 2. The axioms of a loose distributive law

$$\begin{array}{ccccccccccccccc}
TTTA & \xrightarrow{TTM} & TTSA & \xrightarrow{T\delta} & TSTA & \xrightarrow{\delta T} & STTA & \xrightarrow{STM} & STSA & \xrightarrow{S\delta} & SSTA & \xrightarrow{SSM} & SSSA \\
\parallel & & \parallel & & \parallel & & \bar{\delta} & & \parallel & & \parallel & & \parallel \\
TTTA & \xrightarrow{TTM} & TTSA & \xrightarrow{T\delta} & TSTA & \xrightarrow{TSM} & TSSA & \xrightarrow{\delta S} & STSA & \xrightarrow{S\delta} & SSTA & \xrightarrow{SSM} & SSSA \\
\downarrow T\nu & & & \downarrow Tm & & \downarrow T\mu & & \bar{\mu} & & \downarrow \mu T & & \downarrow \mu M & \downarrow \mu S \\
TTA' & \xrightarrow{\quad TM \quad} & TSA & \xrightarrow{\quad \delta \quad} & STA & \xrightarrow{\quad SM \quad} & SSA & \\
\downarrow \nu & & & \downarrow m & & & \downarrow \mu & \\
TA & \xrightarrow{\quad \quad \quad} & SA &
\end{array}$$

(In the last axiom, note that we needed to use the loose pseudonaturality of δ to make the domains match up.)

Definition 7.3. If (A, M, e_M, m_M) and (B, N, e_N, m_N) are δ -monoids, a δ -monoid morphism consists of a tight arrow $f : A \rightarrow B$ and a cell

$$\begin{array}{ccc}
TA & \xrightarrow{M} & SA \\
Tf \downarrow & \bar{f} \Downarrow & \downarrow Sf \\
TB & \xrightarrow{N} & SB
\end{array}$$

such that $\bar{f} \cdot e_M = e_N$ and $\bar{f} \cdot m_M = m_N \cdot (T\bar{f} \bullet \delta_f \bullet S\bar{f})$.

Definition 7.4. If $(f, \bar{f})$ and $(g, \bar{g})$ are δ -monoid morphisms from (A, M, e_M, m_M) to (B, N, e_N, m_N) , a δ -monoid transformation $\alpha : f \Rightarrow g$ consists of a cell

$$\begin{array}{ccc}
A & \xrightarrow{\text{Id}_A} & A \\
\downarrow \iota_B \cdot g & \alpha \Downarrow & \downarrow \eta_B \cdot f \\
TB & \xrightarrow{N} & SB
\end{array}$$

such that the following two composites are equal:

$$\begin{array}{ccc}
\begin{array}{c}
TA \xrightarrow{\quad M \quad} SA \\
\parallel \\
TA \xrightarrow{T\text{Id}_A} TA \xrightarrow{\text{Id}_{TA}} TA \xrightarrow{M} SA \\
\downarrow Tg \quad \downarrow Tf \quad \downarrow \text{Id}_{Tf} \quad \downarrow Tf \quad \downarrow \bar{f} \quad \downarrow Sf \\
TB \xrightarrow{T\alpha} TB \xrightarrow{\text{Id}_{TB}} TB \xrightarrow{N} SB \\
\downarrow T\iota_B \quad \downarrow T\eta_B \quad \downarrow \bar{\eta}_B \quad \downarrow \eta_{TB} \quad \downarrow \eta_N \quad \downarrow \eta_{SB} \\
TTB \xrightarrow{TM} TSB \xrightarrow{\delta_B} STB \xrightarrow{SN} SSB \\
\downarrow \nu_B \quad \quad \quad \downarrow m_N \Downarrow \quad \quad \quad \downarrow \mu_B \\
TB \xrightarrow{\quad N \quad} SB
\end{array}
& \cong &
\begin{array}{c}
TA \xrightarrow{\quad M \quad} SA \\
\parallel \\
TA \xrightarrow{M} SA \xrightarrow{\text{Id}_{SA}} SA \xrightarrow{S\text{Id}_A} SA \\
\downarrow Tg \quad \downarrow \bar{g} \quad \downarrow Sg \quad \downarrow \text{Id}_{Sg} \quad \downarrow Sg \quad \downarrow Sf \\
TB \xrightarrow{N} SB \xrightarrow{\text{Id}_{SB}} SB \xrightarrow{S\alpha} SB \\
\downarrow \iota_{TB} \quad \downarrow \iota_M \quad \downarrow \iota_{SB} \quad \downarrow \bar{\iota}_B \quad \downarrow S\iota_B \quad \downarrow S\eta_B \\
TTB \xrightarrow{TM} TSB \xrightarrow{\delta_B} STB \xrightarrow{SN} SSB \\
\downarrow \nu_B \quad \quad \quad \downarrow m_N \Downarrow \quad \quad \quad \downarrow \mu_B \\
TB \xrightarrow{\quad N \quad} SB
\end{array}
\end{array}$$

Theorem 7.5. δ -monoids, δ -monoid morphisms, and δ -monoid transformations form a 2-category.

Proof. Composition of morphisms is easy: simply compose both the tight arrow and cell components.

We define the postwhiskering of a transformation $\alpha : f \Rightarrow g$ by a morphism $h : N \rightarrow P$ to be $\bar{h} \cdot \alpha$. The transformation axiom follows by postcomposing that for α with $\bar{h}$ and using the morphism axiom $\bar{h} \cdot m_N = m_P \cdot (T\bar{h} \bullet \delta_h \bullet S\bar{h})$, the naturality of ι, η , and the modification/naturality of $\bar{\iota}, \bar{\eta}$. For associativity of postwhiskering, we have $\bar{k} \cdot (\bar{h} \cdot \alpha) = (\bar{k} \cdot \bar{h}) \cdot \alpha = \overline{k \cdot h} \cdot \alpha$, and similarly for identities.

We define the prewhiskering of a transformation $\alpha : f \Rightarrow g$ by a morphism $h : P \rightarrow M$ to be $\alpha \cdot \text{Id}_h$. The transformation axiom follows by precomposing that for α with $\bar{h}$ and using the naturality of the constraints for loose composition and for the pseudofunctoriality of T and S . For associativity we have $(\alpha \cdot \text{Id}_k) \cdot \text{Id}_h = \alpha \cdot (\text{Id}_k \cdot \text{Id}_h) = \alpha \cdot \text{Id}_{k \cdot h}$, and similarly for identities.

The identity transformation on a morphism $f : M \rightarrow N$ is the cell $e_N \cdot \text{Id}_f$. We define the composite of two transformations $\alpha : f \Rightarrow g$ and $\beta : g \Rightarrow h$ to be the following composite (precomposed with a unit isomorphism):

$$\begin{array}{ccccccc}
 A & \xrightarrow{\text{Id}_A} & A & \xrightarrow{\text{Id}_A} & A & \xrightarrow{\text{Id}_A} & A \\
 \downarrow h & & \downarrow g & & \downarrow g & & \downarrow f \\
 B & \xrightarrow{\beta} & B & \xrightarrow{\text{Id}_B} & B & \xrightarrow{\alpha} & B \\
 \downarrow \iota_B & & \downarrow \eta_B & & \downarrow \iota_B & & \downarrow \eta_B \\
 TB & \xrightarrow{N} & SB & \xrightarrow{\bar{\iota} \cdot \text{Id}_\eta = \bar{\eta} \cdot \text{Id}_\iota} & TB & \xrightarrow{N} & SB \\
 \downarrow \iota_{TB} & & \downarrow \iota_N & & \downarrow \eta_{TB} & & \downarrow \eta_N \\
 TTB & \xrightarrow{TN} & TSB & \xrightarrow{\delta_B} & STB & \xrightarrow{SN} & SSB \\
 \downarrow \nu_B & & & & \downarrow m_N \Downarrow & & \downarrow \mu_B \\
 TB & \xrightarrow{\quad\quad\quad} & N & \xrightarrow{\quad\quad\quad} & SB & &
 \end{array}$$

The transformation axiom for this follows by using the associativity axiom for the δ -monoid N and the transformation axioms for α and β . Associativity of this composite similarly follows from associativity for N , while unitality follows from the unit axiom for N . [TODO: I haven't really checked these.]

It remains to show that all three kinds of composition satisfy the interchange law. [Also TODO] $\square$

In the rest of the paper we will define a loose distributive law for which δ -monoids are kits (or, more precisely, “pre-kits”). Of course, first we need a couple of monads. In a kit there are actually *four* different kinds of structure that have to be incorporated into these monads: the extraordinary pairing of variables in the domain, the extraordinary pairing in the codomain, the possibility of multiple modules in the domain, and the substitution of functor terms for abstract variables in the codomain. To simplify the construction and clarify the essential point (composing extraordinary natural transformations), we will first define a loose distributive law for “unary kits without substitution” that includes only the first two; then we will add in the other two.

8. LINEAR DOUBLE MONADS

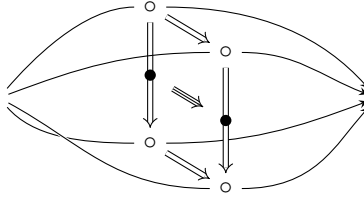
Many of the monads we will be interested in are particularly simple: they are the monads on a slice $\mathbf{Cat}/A$ induced by an internal category in $\mathbf{Cat}$ (that is, a strict double category) with A as object-of-objects. In particular, their underlying functors are given by span composition. Functors between slice categories of this sort are sometimes called **linear** (by analogy with the more general *polynomial* functors that also involve an exponential); they and the “linear” transformations between them are induced by morphisms and 2-cells in the bicategory of spans in $\mathbf{Cat}$.

However, we need not only monads on the ordinary category $\mathbf{Cat}$, but on the double category $\mathbb{C}at$; and also loose distributive laws between these. For this it is convenient to have a structure that describes “spans in $\mathbb{C}at$ ” analogous to the bicategory of spans in the 1-category $\mathbf{Cat}$. There are several ways one could organize such a structure, but for us the important one will be the following.

Let $\mathcal{D}bl$ denote the 2-category of pseudo double categories and strong double functors. This has finite products, so we can regard it as cartesian monoidal.

Definition 8.1 ([?]). A **locally cubical bicategory** is a $\mathcal{D}bl$ -enriched bicategory. Thus, it has objects with hom-double-categories $A(x, y)$, strong composition functors $A(y, z) \times A(x, y) \rightarrow A(x, z)$, strong unit functors $1 \rightarrow A(x, x)$, and associativity and unitality tight natural isomorphisms satisfying the usual axioms.

We call the objects of $A(A, B)$ the **morphisms** of A and write them $A \rightrightarrows B$. We call the tight and loose arrows of $A(A, B)$ **tight 2-cells** and **loose 2-cells** of A , and write them as $f \rightarrow g$ and $f \rightrightarrows g$ respectively, and we call the cells of $A(A, B)$ **3-cells** of A . We refer to composition of tight 2-cells in $A(A, B)$ (and composition of 3-cells in the tight direction) as **transversal**, composition of loose 2-cells in $A(A, B)$ (and composition of 3-cells in the loose direction) as **vertical**, and the action of the composition functors $A(B, C) \times A(A, B) \rightarrow A(A, C)$ as **horizontal**. We have in mind the following picture:



We write transversal composition with $\cdot$ or juxtaposition in applicative order, vertical composition with $\bullet$ in diagrammatic order, and horizontal composition with $\circ$ in applicative order. The unit functors $1 \rightarrow A(A, A)$ pick out an identity morphism id_A for each object A . In addition, every morphism f has an identity tight 2-cell 1_f and an identity loose 2-cell Id_f , every tight 2-cell α has an identity 3-cell Id_α , every loose 2-cell β has an identity 3-cell 1_β , and we have $\text{Id}_{1_f} = 1_{\text{Id}_f}$. We have $\text{Id}_f \circ \text{Id}_g \cong \text{Id}_{f \circ g}$ and $1_g \circ 1_f = 1_{f \circ g}$, as well as isomorphisms

$$(8.2) \quad (p \circ q) \bullet (r \circ s) \cong (p \bullet r) \circ (q \bullet s).$$

Note that the objects, morphisms, and tight 2-cells of a locally cubical bicategory A form an ordinary bicategory under transversal and horizontal composition, which we denote $\mathcal{H}(A)$. On the other hand, a locally cubical bicategory can be regarded as

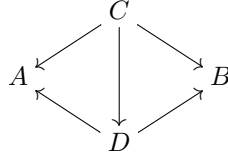
a degenerate sort of *intercategory* [?]. We will not need this point of view very much, but our notation is chosen to match theirs (except that we write $\circ$ in functional rather than diagrammatic order).

There are two main examples we care about.

Example 8.3. The cartesian monoidal 2-category $\mathcal{Dbl}$ is in fact closed as a monoidal bicategory, i.e. the functors $A \times -$ have right biadjoints. As observed in [?], it can therefore be regarded as a bicategory enriched over itself, i.e. a locally cubical bicategory. The hom-double-category $\mathcal{Dbl}(A, B)$ consists of strong double functors, tight transformations, loose transformations, and modifications between them.

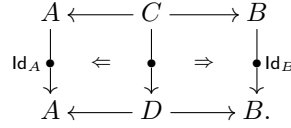
Theorem 8.4. *There is a locally cubical bicategory $\mathcal{L}(\mathbf{Cat})$ described as follows.*

- *Its objects are groupoids.*
- *Its morphisms are spans $A \leftarrow C \rightarrow B$ in $\mathbf{Cat}$ such that A and B are groupoids and $C \rightarrow A$ is a fibration. Composition is by pullback, which is well-defined since fibrations are closed under pullback and composition.*
- *Its tight 2-cells are span morphisms:*



Thus the underlying bicategory is a sub-bicategory of $\mathcal{Span}(\mathbf{Cat})$.

- *Its loose 2-cells from $A \leftarrow C \rightarrow B$ to $A \leftarrow D \rightarrow B$ are profunctors $C \twoheadrightarrow D$ equipped with a span of 2-cells in $\mathbf{Cat}$:*



These are composed by pullback in $\mathbf{Cat}_1$.

- *Its 3-cells are span morphisms in $\mathbf{Cat}_1$.*

Proof. Recall from [?, ?] that an “intercategory” is an internal category in the 2-category $\mathcal{LxDb}$ of double categories and *lax* double functors whose source and target functors are strict. As observed in [?, 3.5], a locally cubical bicategory is an intercategory whose double category of objects is discrete and whose composition and unit functors are strong.

If an intercategory has strictly unital vertical composition (which can usually — though not always — be obtained by redefining vertical composition at identities), then we can discard all the arrows and cells in its double category of objects to obtain another intercategory with a discrete double category of objects. Applying this to the example of [?, 6.5], we get a structure that is almost like a locally cubical bicategory, but its composition functors are only lax normal rather than strong (though its unit functor is strong). However, when we restrict the objects to groupoids and the spans to have one leg a fibration, then the composition functor becomes strong as well, by [?, Proposition 49]. $\square$

Now, a linear functor $\mathbf{C}/A \rightarrow \mathbf{C}/B$ induced by a span $A \leftarrow C \rightarrow B$ in $\mathbf{C}$ can be described in terms of the bicategory $\mathcal{Span}(\mathbf{C})$, by identifying $\mathbf{C}/A$ with the

hom-category $\mathcal{S}pan(\mathbf{C})(1, A)$ and composing them with the given span $A \leftarrow C \rightarrow B$. We can do something similar in $\mathcal{L}(\mathbf{Cat})$, where the slice double category $\mathbf{Cat}/A$ can be identified with the hom-double-category $\mathcal{L}(\mathbf{Cat})(1, A)$. The following observation is entirely straightforward.

Theorem 8.5. *Any object X in a locally cubical bicategory $\mathbf{A}$ induces a representable locally cubical pseudofunctor $\mathbf{A}(X, -) : \mathbf{A} \rightarrow \mathbf{Dbl}$. $\square$*

In particular, any “horizontal monad” on $A \in \mathbf{A}$ (whose multiplication and unit are tight 2-cells) induces a strong double monad on any hom-double-category $\mathbf{A}(X, A)$. Taking $\mathbf{A} = \mathcal{L}(\mathbf{Cat})$ and $X = 1$, we see that *any linear monad on $\mathbf{Cat}/A$ induces a strong double monad on $\mathbf{Cat}/A$* . (In a later paper we will show that something similar is true for all polynomial monads.)

The punchline is that loose distributive laws are *essentially* a “locally cubical” concept. If literally true, this would mean they could be defined in any locally cubical bicategory and are preserved by locally cubical pseudofunctors. To see this, simply take ?? and replace functors by 1-cells, tight and loose transformations by tight and loose 2-cells, square modifications by square 3-cells, and so on. We chose our notations for composition so that the pictures would require no changes; the pseudonaturality of δ is replaced by the interchanger (??).

The reason this is not *exactly* true is that in stating ?? we used the fact that composition of pseudo double functors is strictly associative and unital, whereas in a general locally cubical bicategory, $\circ$ is only associative and unital up to isomorphism. One is tempted to say “just insert the associativity and unitality constraints everywhere”, but this doesn’t work because these are *tight* 2-cells, whereas in the domain of (say) $\bar{\mu}$ we would need an associativity to appear in the *middle* of a composite of two *loose* isomorphisms.

The solution to this is fairly standard in double-category theory, and goes by various names such as “fibrancy”, “framing”, or “transportability”: we require that the relevant tight 2-cells have *loose companions*. [TODO: Define.]

With this modification, we *can* define a loose distributive law in any fibrant locally cubical bicategory, and they will be preserved by any locally cubical functor (since any strong double functor preserves companions). This applies in particular to the functor $\mathcal{L}(\mathbf{Cat})(1, -)$, so that to define a loose distributive law between monads on a slice double category $\mathbf{Cat}/A$, it suffices to define a corresponding structure on A in $\mathcal{L}(\mathbf{Cat})$. The latter is a combinatorial structure amenable to concrete description.

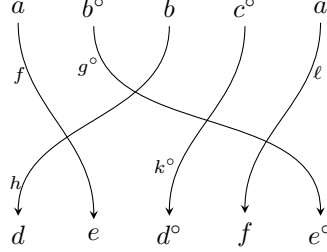
9. EXTRAORDINARY NATURALITY

Let V be the monad on $\mathbf{Cat}$ whose algebras are symmetric strict monoidal categories equipped with a strict symmetric monoidal strict involution, i.e. a strict symmetric monoidal functor $(-)^{\circ} : \mathcal{C} \rightarrow \mathcal{C}$ such that $(a^{\circ})^{\circ} = a$ functorially. Concretely, the objects of $V\mathcal{C}$ are finite lists of objects of $\mathcal{C}$, some of which are marked with a formal $(-)^{\circ}$, such as $(a, b^{\circ}, b, c^{\circ}, a)$. It will also sometimes be convenient to write such a list as $(a^{+}, b^{-}, b^{+}, c^{-}, a^{+})$, where $a^{+} = a$ and $a^{-} = a^{\circ}$; we can then write a^{ε} to mean either a^{+} or a^{-} depending on whether $\varepsilon = +$ or $\varepsilon = -$. We write ε^{*} for the reversed variance, i.e. $+^{*} = -$ and $-^{*} = +$.

Since $V\mathcal{C}$ is monoidal with an involution, its objects can be concatenated and oppositized (by distributing the opposite over the list), so for instance

$$(a, b^{\circ})(c^{\circ}, d, a) = (a, b^{\circ}, c^{\circ}, d, a) \quad \text{and} \quad (b^{\circ}, a, a)^{\circ} = (b, a^{\circ}, a^{\circ}).$$

Its unit object is $\mathbb{1} = ()$. The morphisms of VC are given by finite lists of morphisms of $\mathcal{C}$ labeled by permutations which respect the opposites, e.g. a morphism $(a, b^\circ, b, c^\circ, a) \rightarrow (d, e, d^\circ, f, e^\circ)$ might be:



Here $f : a \rightarrow e$, $g : b \rightarrow e$, $h : b \rightarrow d$, $k : c \rightarrow d$, and $\ell : a \rightarrow f$ are morphisms in $\mathcal{C}$. Note that there can only be a morphism between two lists if they have the same length *and* the same number of opposites. Formally, if $A = (a_1^{\varepsilon_1}, \dots, a_n^{\varepsilon_n})$ and $B = (b_1^{\varphi_1}, \dots, b_n^{\varphi_n})$, then a morphism $A \rightarrow B$ is a pair (σ, f) where $\sigma \in \Sigma_n$ is a permutation such that $\varphi_{\sigma(k)} = \varepsilon_k$, and $f = (f_1, \dots, f_n)$ is a list of morphisms with $f_n : a_n \rightarrow b_{\sigma(n)}$.

For the rest of this section, let $\mathcal{C}$ be a *discrete* category, whose objects will be the cats of our unary kit. In particular, therefore VC is a groupoid. We will define a loose distributive law on VC in $\mathcal{L}(\mathbf{Cat})$, inducing one on $\mathbf{Cat}/VC$ whose generalized polycategories are “unary kits without functors”.

The two monads involved in this distributive law will in fact be the same monad, which we denote R . Being a horizontal monad in $\mathcal{L}(\mathbf{Cat})$, it is equivalently a monad in the bicategory $\mathbf{Span}(\mathbf{Cat})$, i.e. an internal category in $\mathbf{Cat}$, i.e. a *strict* double category $\mathbb{R}$. Since its category of objects $\mathbb{R}_0$ is VC , its objects are those of VC and its tight morphisms are the morphisms of VC . We now define its loose morphisms and cells.

Let us call an object of VC **totally covariant** if it contains no $(-)^{\circ}$'s. A **B -trivial pre-matching** between $A, B \in VC$ is an isomorphism $A \cong BEE^{\circ}$ in VC where E is totally covariant and the permutation preserves the relative order of cats in B . Two pre-matchings are **equivalent** if they are related by an isomorphism $E \cong E'$ (necessarily unique if it exists); a **B -trivial matching** is an equivalence class of B -trivial pre-matchings. [Is this passage to equivalence classes the right thing to do? Does it actually make sense everywhere?]

Now a loose morphism from A to B in our double category (an object of $\mathbb{R}_1$) is a B -trivial matching between A and B . The loose composite of matchings $A \cong BDD^{\circ}$ and $B \cong CEE^{\circ}$ is the composite

$$A \cong BDD^{\circ} \cong CEE^{\circ}DD^{\circ} \cong C(ED)(ED)^{\circ}.$$

The identity matching is the unique one with $E = ()$. The cells are commutative squares of two matchings and two isomorphisms, in an obvious sense.

This completes the definition of a strict double category $\mathbb{R}$ with $\mathbb{R}_0 = VC$. Note that the source and target functors of $\mathbb{R}$ are both individually fibrations (though the functor $(s, t) : \mathbb{R}_1 \rightarrow \mathbb{R}_0 \times \mathbb{R}_0$ is not a fibration). Thus, in particular it is a monad in $\mathcal{L}(\mathbf{Cat})$, and thus induces a strong double monad R on $\mathbf{Cat}/VC$.

An object of $\mathbf{Cat}/VC$ is a functor $\mathcal{E} \rightarrow VC$, where the fiber $\mathcal{E}(A)$ over $A = (a_1^{\varepsilon_1}, \dots, a_n^{\varepsilon_n})$ is the set of modules with boundary A . Usually we will assume it to be a discrete fibration (or equivalently a discrete opfibration, since VC is a

groupoid), meaning that we can permute the cats in the boundary of a module. The objects of $R\mathcal{E}$ are then obtained by choosing an object of $\mathcal{E}$, pairing up some of the cats in its boundary into covariant and contravariant pairs, and removing them from the boundary of the result. Note that the passage from pre-matchings to matchings means that there is no ordering imposed on the paired cats, and B -triviality means that the order of the unremoved cats is maintained. However, $R\mathcal{E} \rightarrow VC$ is still no longer a *discrete* fibration: an object in the fiber over B consists of a B -trivial matching $A \cong BEE^\circ$ together with an object of $\mathcal{E}(A)$, and if E admits automorphisms in VC then so can such objects.

Now a profunctor $\mathcal{M} : R\mathcal{E} \multimap R\mathcal{E}$ in $\mathbf{Cat}/VC$ consists of a set of “morphisms”, each of which is assigned a pair of modules with some of their cats paired up to be its domain and codomain, plus a permutation relating the *unpaired* cats in the domain and the codomain. The latter datum is the 2-cell

$$\begin{array}{ccc} R\mathcal{E} & \longrightarrow & VC \\ \mathcal{M} \bullet \downarrow & \Rightarrow & \bullet \downarrow \text{Id}_{VC} \\ R\mathcal{E} & \longrightarrow & VC \end{array}$$

exhibiting $\mathcal{M}$ as a loose morphism in $\mathbf{Cat}/VC$. These elements of $\mathcal{M}$ are the extraordinary morphisms between modules, with the pairings in their domain and codomain representing the extraordinary-natural cats. The profunctorial action means that the cats in the domains and codomains of morphisms can be permuted, including the extraordinary natural ones.

Now we have to define a loose distributive law of R over itself on $\mathbf{Cat}/VC$. As suggested in ??, we will do this by defining a loose distributive law of $\mathbb{R}$ over itself in the locally cubical bicategory $\mathcal{L}(\mathbf{Cat})$. For clarity, we write $\mathbb{S} = \mathbb{T} = \mathbb{R}$, with $\mathbb{S}$ as a mnemonic for source and $\mathbb{T}$ for target; thus our goal is $\delta : \mathbb{T} \circ \mathbb{S} \multimap \mathbb{S} \circ \mathbb{T}$.

Now $\mathbb{T} \circ \mathbb{S}$ and $\mathbb{S} \circ \mathbb{T}$ are spans from VC to itself, whose objects consist of a B -trivial matching $A \cong BDD^\circ$ and a C -trivial matching $B \cong CEE^\circ$, with the projections selecting A and C . Of course these have a composite C -trivial matching from A to C ; the additional data beyond the composite is simply a division of the paired cats into those in D and those in E . The object $B \in VC$ is actually determined once we know D and the composite, by B -triviality of the first matching. In particular, the fibers of $\mathbb{T} \circ \mathbb{S} \rightarrow VC \times VC$ are discrete.

Our δ is supposed to be a profunctor $\mathbb{T} \circ \mathbb{S} \multimap \mathbb{S} \circ \mathbb{T}$, which moreover must be a loose 2-cell in $\mathcal{L}(\mathbf{Cat})$ and thus come with cells

$$\begin{array}{ccccc} VC & \longleftarrow & \mathbb{T} \circ \mathbb{S} & \longrightarrow & VC \\ \text{Id}_{VC} \bullet \downarrow & \Leftarrow & \bullet \downarrow & \Rightarrow & \bullet \downarrow \text{Id}_{VC} \\ VC & \longleftarrow & \mathbb{S} \circ \mathbb{T} & \longrightarrow & VC. \end{array}$$

Taken together, this means that for any A, C and A', C' , and isomorphisms $A \cong A'$ and $C \cong C'$, plus a C -trivial matching from A to C and a C' -trivial matching from A' to C' , both with the paired cats partitioned into two subsets, we must have a specified “set of ways to compose”. Put differently, the “type” of an element of δ is

given by the following data:

$$(9.1) \quad \begin{array}{ccccc} A & \xrightarrow{\cong} & BDD^\circ & \xrightarrow{\cong} & CEE^\circ DD^\circ & & C \\ \cong \downarrow & & & & & & \downarrow \cong \\ A' & \xrightarrow{\cong} & B'D'(D')^\circ & \xrightarrow{\cong} & C'E'(E')^\circ D'(D')^\circ & & C' \end{array}$$

Note that the isomorphism $C \cong C'$ is not required to “commute” with the others.

As the first step in our definition, we specify that any such “type” will be inhabited by *at most one* element of δ . It is simplest to describe the conditions under which this happens as a partial function from the left half of the diagram $BDD^\circ \cong A \cong A' \cong B'D'(D')^\circ$ to the right half. Regard this composite isomorphism $BDD^\circ \cong B'D'(D')^\circ$ as a bipartite graph, and add additional edges connecting each cat in D or D' to its counterpart in D° or $(D')^\circ$, respectively. In the resulting graph all vertices have degree 1 or 2, so it is a disjoint union of paths and cycles. The partial function is defined if and only if there are *no* cycles. In this case, each path must start and end at a cat in B or B' . Every path that connects two cats in B gives us an element of E , while every path that connects two cats in B' gives us an element of E' . The remaining paths connect one cat in B with one in B' , specifying the isomorphism $C \cong C'$. Note that the variance gets flipped every time a path “changes direction”, so E - and E' -paths do indeed connect two copies of a cat with opposite variance, while the remaining ones preserve variance.

To be a profunctor, δ must also be acted on by permutations on both sides, compatibly with the projections to $\mathbf{Id}_{VC}$ on both sides; but this is evident. Thus, we have defined a loose 2-cell $\delta : \mathbb{T} \circ \mathbb{S} \rightarrow \mathbb{S} \circ \mathbb{T}$ in $\mathcal{L}(\mathbf{Cat})$.

For the remaining data and axioms of a loose distributive law from ??, we note that $\mathbb{T}$ and $\mathbb{S}$ play entirely symmetrical roles in the construction. Thus, we can cut our work almost in half by symmetry (“almost” because two of the axioms are self-dual under this symmetry).

[TODO ...]

10. SUBSTITUTIONS

Next we incorporate nontrivial functors into our unary kits. We start by describing the structure formed by the cats and functors as a kind of generalized multicategory in its own right.

We extend V to a monad on the double category $\mathbf{Cat}$ in the usual way (in fact, the unique way), by treating the elements of profunctors as if they were “morphisms”. That is, the above diagram could also represent an element of a profunctor $VH : V\mathcal{D} \rightarrow V\mathcal{C}$, if $a, b, c \in \mathcal{C}$ and $d, e, f \in \mathcal{D}$ while $f \in H(a, e)$, $g \in H(b, e)$, and so on. The technology of generalized multicategories then yields a notion of *virtual V -algebra*, which contains “multimorphisms” whose domain is a list with variance like $(a, b^\circ, b, c^\circ, a)$. Note that these are like symmetric multicategories in that symmetric groups act on the domain lists.

As usual, any virtual V -algebra $\mathcal{C}$ generates a free V -algebra $\widehat{\mathcal{C}}$, whose objects are the lists with variance of objects in $\mathcal{C}$, and whose morphisms are lists of multimorphisms in $\mathcal{C}$, with domain and codomain arbitrarily permuted. More precisely, a morphism from $A = (a_1^{\varepsilon_1}, \dots, a_n^{\varepsilon_n})$ to $B = (b_1^{\varphi_1}, \dots, b_m^{\varphi_m})$ in $\widehat{\mathcal{C}}$ is represented by a pair (σ, f) where σ is a bijection $(\sum_{i=1}^m k_i) \xrightarrow{\sim} n$ and $f = (f_1, \dots, f_m)$ is a list of

morphisms $f_i : (a_{\sigma(i,1)}^{\varphi_i \varepsilon_{\sigma(i,1)}}, \dots, a_{\sigma(i,k_i)}^{\varphi_i \varepsilon_{\sigma(i,k_i)}}) \rightarrow b_i$. (Here $\varphi_i \varepsilon_{\sigma(i,j)}$ means the usual multiplication of signs.) We quotient these pairs by the inner symmetric group actions, i.e. given permutations $\tau_i \in \Sigma_{k_i}$ we have $(\sigma(\sum_i \tau_i), f) = (\sigma, f\tau)$, where $(f\tau)_i = f_i \tau_i$ is defined by the symmetric action on the domains of morphisms in $\mathcal{C}$. More formally, the hom-profunctor of $\widehat{\mathcal{C}}$ is given by $\text{hom}_{\widehat{\mathcal{C}}} = V \text{hom}_{\mathcal{C}} \otimes_{VVC} V\mathcal{C}(1, \nu)$. Since the monoidal structure on $\widehat{\mathcal{C}}$ is induced by the concatenation monoidal structure on $V\mathcal{C}$, we also write it as juxtaposition.

Note that every isomorphism in $\widehat{\mathcal{C}}$ consists only of unary arrows. By a **permutation isomorphism** in $\widehat{\mathcal{C}}$ we mean an isomorphism in which these unary arrows are all identities; thus it really is nothing but a permutation. In fact, the subcategory of permutation isomorphisms in $\widehat{\mathcal{C}}$ is isomorphic to $V\mathcal{C}_0$, where $\mathcal{C}_0$ is the set of objects of $\mathcal{C}$.

Now fix a virtual V -algebra $\mathcal{C}$, whose objects and arrows will be our cats and functors. We define a new linear double monad incorporating restriction along morphisms in $\mathcal{C}$ as follows. Say that a **pre-matching with restriction** between $A, B \in V\mathcal{C}$ is a morphism $BEE^\circ \rightarrow A$ in $V\mathcal{C}$ where E is totally covariant. Two such pre-matchings are **equivalent** if they are related by an isomorphism $E \cong E'$ (necessarily unique if it exists); a **matching with restriction** is an equivalence class of pre-matchings.

Now let $\mathbb{C}$ be the strict double category whose tight morphisms are the permutation isomorphisms (that is, $\mathbb{C}_0 = V\mathcal{C}_0$), whose loose morphisms from A to B are matchings with restrictions from B to A , and whose cells are commutative squares in $\widehat{\mathcal{C}}$. [Hmm, does this make sense with the passage to equivalence classes?] Then $\mathbb{C}$ is a monad in $\mathcal{L}(\text{Cat})$ on $V\mathcal{C}_0$. We will construct a loose distributive law $\mathbb{C} \circ \mathbb{S} \rightarrow \mathbb{S} \circ \mathbb{C}$ in $\mathcal{L}(\text{Cat})$, analogous to the one constructed in the previous section.

As a loose 2-cell in $\mathcal{L}(\text{Cat})$, this will look like

$$\begin{array}{ccccc} V\mathcal{C}_0 & \longleftarrow & \mathbb{C} \circ \mathbb{S} & \longrightarrow & V\mathcal{C}_0 \\ \text{Id}_{V\mathcal{C}_0} \downarrow & & \Downarrow & & \downarrow \text{Id}_{V\mathcal{C}_0} \\ V\mathcal{C}_0 & \longleftarrow & \mathbb{S} \circ \mathbb{C} & \longrightarrow & V\mathcal{C}_0. \end{array}$$

Now the “type” of an element of δ is given by the following data, in which some of the isomorphisms from $(??)$ have been replaced by morphisms in $\widehat{\mathcal{C}}$:

$$\begin{array}{ccc} A \xrightarrow{\cong} BDD^\circ \longleftarrow CEE^\circ DD^\circ & & C \\ \cong \downarrow & & \downarrow \cong \\ A' \longleftarrow B'D'(D')^\circ \xrightarrow{\cong} C'E'(E')^\circ D'(D')^\circ & & C' \end{array}$$

As before, any such “type” will be inhabited by at most one element of δ , and we can describe the conditions under which this happens as a partial function from the left half of the diagram to the right half. Thus, suppose given the left half, including in particular a composite morphism $B'D'(D')^\circ \rightarrow A' \cong A \cong BDD^\circ$ in $\widehat{\mathcal{C}}$. By definition of $\widehat{\mathcal{C}}$, this consists of a morphism $f_b : F_b \rightarrow b$ of $\mathcal{C}$ for each $b \in B$ and two morphisms $g_d : G_d \rightarrow d$ and $h_d : H_d \rightarrow d$ for each $d \in D$, together with an isomorphism

$$(10.1) \quad B'D'(D')^\circ \xrightarrow{\sim} F_{b_1}^{\varepsilon_1} \cdots F_{b_n}^{\varepsilon_n} G_{d_1} \cdots G_{d_m} H_{d_1}^\circ \cdots H_{d_m}^\circ$$

The conditions are then:

- (i) $g_d = h_d$ for each d (and in particular, $G_d = H_d$).
- (ii) Regard the isomorphism (??) as a bipartite graph, and add additional edges connecting each cat in D to its counterpart in D° , and each cat in each G_d to its counterpart in H_d° (which is equal to G_d°). Again this is a disjoint union of paths and cycles, and we require that there are no cycles.

When these conditions hold, each path in the above graph must start and end at a cat in B' or some F_b . Every path that connects two cats in B' gives us an element of E' , and we define C' to be the remaining cats in B' , giving the matching $B' \cong C'E'(E')^\circ$. Every path that connects two cats in some F_b 's (perhaps for different b 's) gives us an element of E , and we define C to be the remaining cats in any of the F_b 's. This yields an isomorphism $CEE^\circ \cong F_{b_1}^{\varepsilon_1} \cdots F_{b_n}^{\varepsilon_n}$, and we define the morphism $CEE^\circ \rightarrow B$ to be given by this isomorphism together with the f_b 's. Finally, the remaining paths in the graph connect one cat in B' with one in some F_b , specifying the isomorphism $C \cong C'$. As before, the actions of permutation isomorphisms are trivial, so we have a loose 2-cell in $\mathcal{L}(\text{Cat})$.

[TODO...]

[NOTE: the action of permuting cats in the boundary of a module seems to be getting represented in two different ways here. We start with a category over the groupoid VC_0 of permutation isomorphisms, which seems like it ought to be a discrete fibration. But then we also incorporate formal restrictions along arbitrary morphisms of $\widehat{\mathcal{C}}$ in the codomains of module morphisms, which includes permutation isomorphisms as well, and can then characterize actual restrictions/substitutions by a universal property. How are these two representations related? It seems that we do need to use VC_0 , rather than its discrete set of objects, as the base, since the “composition of graphs” partial function wants to output a permutation.]

11. MULTIVARIABLE MORPHISMS

Finally, we incorporate the possibility of module morphisms with multiple modules in the domain. This takes us out of the realm of linear functors (though polynomial functors would suffice), so we use a different technique. Let P denote the monad on Cat for symmetric strict monoidal categories. With V and $\mathcal{C}$ as in the previous section VC_0 is a P -algebra. Therefore, P induces a monad on Cat/VC_0 (this is true for any monad algebra and any comma object in a 2-category), which we also denote P . We claim that this monad P is connected to the monad S (a.k.a. R and T) defined in ?? by a *tight* distributive law $\sigma : PS \rightarrow SP$.

For a category $\mathcal{E}$ over VC_0 , the objects of $PS\mathcal{E}$ are finite lists $(M_1, \dots, M_n)$ of objects of $S\mathcal{E}$, each of which is an object $M_i \in \mathcal{E}(A_i)$ where $A_i \in VC_0$ is equipped with a B_i -trivial matching $A_i \cong B_i E_i E_i^\circ$. Such an object sits over $B_1 \cdots B_n$ in VC_0 .

On the other hand, the objects of $SP\mathcal{E}$ are finite lists $(M_1, \dots, M_n)$ of objects $M_i \in \mathcal{E}(A_i)$, equipped with a B -trivial matching $A_1 \cdots A_n \cong BEE^\circ$. Thus σ on objects has merely to concatenate the matchings and rearrange them, defining $B = B_1 \cdots B_n$ and $E = E_1 \cdots E_n$:

$$A_1 \cdots A_n \xrightarrow{\sim} B_1 E_1 E_1^\circ \cdots B_n E_n E_n^\circ \xrightarrow{\sim} (B_1 \cdots B_n)(E_1 \cdots E_n)(E_1 \cdots E_n)^\circ$$

The action of σ on morphisms and profunctors is analogous, and its naturality and compatibility with the monad structures of P and S is straightforward.

We also have a similar tight distributive law $\tau : PC \rightarrow CP$, where C is the monad on $\widehat{\mathbf{Cat}}/VC_0$ from ???. The only difference is that we have to concatenate morphisms of $\widehat{C}$ rather than merely permutation isomorphisms.

We intend to assemble these two tight distributive laws with the loose distributive law $\delta : CS \rightarrow SC$ into a loose distributive law $C(SP) \rightarrow (SP)C$, which will finally be the one that characterizes pre-kits as generalized polycategories. The basic idea is contained in [?] where it is shown that given three monads T_1, T_2, T_3 and distributive laws $\lambda_{ij} : T_i T_j \rightarrow T_j T_i$ for $i > j$ satisfying the Yang-Baxter equation

$$\begin{array}{ccccc}
 & & T_2 T_3 T_1 & \xrightarrow{T_2 \lambda_{31}} & T_2 T_1 T_3 \\
 & \nearrow \lambda_{32} T_1 & & & \searrow \lambda_{21} T_3 \\
 T_3 T_2 T_1 & & & & T_1 T_2 T_3 \\
 & \searrow T_3 \lambda_{21} & & & \nearrow T_1 \lambda_{32} \\
 & & T_3 T_1 T_2 & \xrightarrow{\lambda_{31} T_2} & T_1 T_3 T_2
 \end{array}$$

there is (among other things) a composite distributive law $T_3(T_1 T_2) \rightarrow (T_1 T_2)T_3$. Our present situation is complicated by the fact that two of our distributive laws are tight and one is loose. Moreover, just as in the definition of loose distributive laws, instead of a *commutative* Yang-Baxter diagram it is more natural to ask for a (not necessarily invertible) “Yang-Baxterator” modification satisfying its own laws. This leads us to the following definition.

Definition 11.1.

[TODO...]